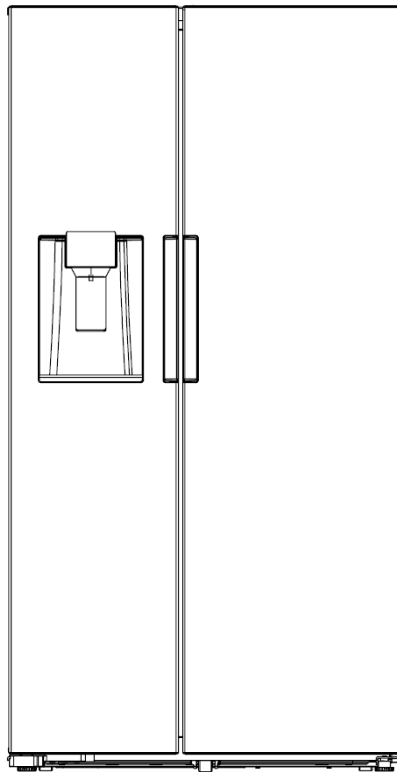


INSIGNIA®

Service Manual

SBS SERIES

Customer model	Factory model
NS-RSS26WH0 NS-RSS26SS0	UR-BCD746WE-DT



The picture in this service manual is only for reference, and specific appearance and configuration are subject to the real product.

This manual mainly teaches the method, the specific work skill needs engineer to accumulate through the daily work.

 **WARNING**

Important Safety Notice

There are special components used in this equipment which are important for safety. These parts are marked by  in the Schematic Diagrams, Circuit Board Diagrams, Exploded Views and Replacement Parts List. It is essential that these critical parts should be replaced with manufacturer's specified parts to prevent shock, fire or other hazards. Do not modify the original design without permission of manufacturer.

WARNING

Important Safety Notice

The Maintenance Manual is only for the use of maintenance personnel with certain experience and background in electrical, electronic and mechanical field.

Any attempt to repair main devices may lead to personal injury and property loss.

Manufacturers or distributors are not responsible for the content of the Manual and interpretation thereof.

Midea Refrigerators

Technical Maintenance Manual

Copyright @2017

All rights reserved. Replication of all or part of the Manual in any forms shall not be allowed without written approval by the Overseas Sales Corporation of Midea Refrigerators.

Contents

1. SIGNIFICANT UPDATE NOTES(NONE)	5
2. SAFETY WARNING CODE	6
2.1 WARNING FOR OPERATION SAFETY	6
2.2 SAFETY INSTRUCTION FOR REFRIGERANT	8
3. TRANSPORT	9
3.1 HANDLING	9
4. INSTALLATION AND COMMISSIONING	10
4.1 DOOR DISASSEMBLY AND ASSEMBLY	10
4.2 INSTALLATION LOCATION	13
4.3 LEVELING OF THE REFRIGERATOR	13
4.4 LEFT OR RIGHT OPEN DOOR REVERSAL(NONE)	14
4.5 INSTALLATION OF HANDLE(NONE)	14
4.6 INSTALLATION OF DOOR LOCK (NONE)	14
4.7 ADJUSTMENT TO LEVEL THE DOOR	14
5. PRODUCT CONFIGURATION AND DIMENSION	16
5.1 MAIN PARTS AND THEIR NAMES	16
5.2 EXTERNAL DIMENSION	17
5.3 LOCATION OF S/N	18
6. PRODUCT SPECIFICATION	19
6.1 ELECTRICAL PARAMETERS	19
6.2 COOLING TEMPERATURE SETTING RANGE	20
6.3 DEFROSTING PARTS	20
6.4 CIRCUIT DIAGRAM	20
6.5 MAIN CONTROL BOARD STRUCTURE DIAGRAM	21
6.5.1 ICE-MAKER CONTROL PANEL	21
6.6 ELECTRIC EXCHANGE VALVE (NONE)	21
6.7 WATER VALVE	22
7. REFRIGERATING PIPING SYSTEM AND CIRCULATING ROUTE OF COOLING AIR	23
7.1 REFRIGERATING PIPING SYSTEM	23
7.2 SPECIAL SOLDERING POSITION (NONE)	24
7.3 CIRCULATING ROUTE OF COOLING AIR	24
8. DISMANTLING OF PARTS	25
8.1 PARTS ON THE DOOR	25
8.2 PARTS INSIDE THE REFRIGERATOR	25
8.3 LIGHT SYSTEM	26
8.4 AIR DUCT COMPONENTS REFRIGERATING CHAMBER AND FAN MOTOR	27
8.5 AIR DUCT COMPONENTS IN FREEZING CHAMBER AND FAN MOTOR	28

8.6 EVAPORATOR AND DEFROST SYSTEM.....	29
8.7 COMPRESSOR CASE.....	31
8.8 DISPLAY CONTROL BOARD	36
8.9 MAIN CONTROL BOARD.....	37
8.10 BAR COUNTER(NONE)	38
8.11 WATER DISPENSER	38
8.12 ICE MAKER.....	38
8.13 WATER FILTER COMPONENT	39
9. TEMPERATURE SENSING SYSTEM.....	41
9.1 POSITION OF SENSORS	41
9.2 REPLACEMENT OF SENSORS	41
9.3 SENSOR REPLACEMENT	43
9.4 SENSOR RESISTANCE(R/T).....	44
10. FUNCTION AND OPERATION.....	45
10.1 OPERATION PANEL	45
10.2 DISPLAY	46
10.3 TEMPERATURE CONTROL.....	46
10.4 OPEN DOOR ALARM.....	47
10.5 FUNCTION SELECTION	47
10.6 CONTROL OF ICE MAKER.....	47
10.7 CELSIUS FAHRENHEIT CONVERSION	47
10.8 FAULT CODE AND SOLUTIONS	47
10.9 DEFROSTING FUNCTION.....	50
10.10 EYE ICE MACHINE TEST MODE.....	50
10.11 BACKUP DATA FOR POWER FAIL	51
11. COMPRESSOR.....	52
11.1 COMPRESSOR ON AND OFF CONTROL SPECIFICATIONS	52
11.2 VARIABLE FREQUENCY DRIVER BOARD FAULT ANALYSIS	52
12. TROUBLESHOOTING METHOD.....	53
12.1 NO COOLING(AIR COOLING-ELECTRONIC)	53
12.2 NO WORKING OF COMPRESSOR.....	54
12.3 INSIDE FROSTING, NO DEFROSTING	54
12.4 INSIDE FROSTING, NO DEFROSTING-MAINTENANCE GUIDELINES.....	55
12.5 LIGHT IS NOT ON.....	56
12.6 FAN FAILURE	56
12.7 DEFECTIVE DEFROST CIRCUIT	57
12.8 NOISE.....	57
12.9 AIR DUCT NOT OPERATED(ELECTRONICALLY).....	58
12.10 ICE MAKER NOT MAKE ICE	58

1. Significant update notes(None)

2. Safety Warning Code

2.1 Warning for operation safety

Important Safety Instructions

	CAUTION RISK OF ELECTRIC SHOCK DO NOT OPEN	
	This symbol indicates that dangerous voltage constituting a risk of electric shock is present within your freezer.	
	This symbol indicates that there are important operating and maintenance instructions in the literature accompanying your freezer.	

WARNING

- 1) Read these instructions.
- 2) Keep these instructions.
- 3) Heed all warnings.
- 4) Follow all instructions.
- 5) Do not use this appliance near water.
- 6) Clean only with a damp cloth.
- 7) Do not block any ventilation openings.
- 8) Install in accordance with the manufacturer's instructions.
- 9) Do not install near any heat sources, such as radiators, heat registers, stoves, or other apparatus that produce heat.
- 10) Do not defeat the safety purpose of the polarized or grounding-type plug. A polarized plug has two blades with one wider than the other. A grounding type plug has two blades and a third grounding prong. The wide blade or the third prong are provided for your safety. If the provided plug does not fit into your outlet, consult an electrician for replacement of the obsolete outlet.
- 11) Protect the power cord from being walked on or pinched, particularly at plugs, convenience receptacles, and the point where they exit from the appliance.
- 12) Do not attempt to modify or extend the power cord of this appliance.
- 13) Unplug this appliance during lightning storms or when it will not be used for long periods of time.
- 14) Make sure that the available AC power matches the voltage requirements of this appliance.

CONNECTING ELECTRICITY

**WARNING****Electrical Shock Hazard.**

Plug into a grounded 3-prong outlet.
Do not remove the ground prong.
Do not use an adapter.

Failure to follow these instructions can result in death, fire, or electrical shock.

**WARNING****Electric Shock Hazard**

Failure to follow these instructions can result in electric shock, fire, or death.

- 1) **WARNING**—Keep ventilation openings, in both the freezer and the built-in structure, clear of obstruction.
- 2) **WARNING**—Do not touch the interior of the freezer with wet hands. This could result in frost bite.
- 3) **WARNING**—Do not use mechanical devices or other means to accelerate the defrosting process, other than those recommended by the manufacturer.
- 4) **WARNING**—Do not damage the refrigerant circuit.
- 5) **WARNING**—Do not damage the refrigerant tubing when handling, moving, or using the freezer.
- 6) **WARNING—DANGER**—Never allow children to play with, operate, or crawl inside the freezer. Risk of child entrapment. Before you throw away your old freezer:
 - 6-1) Take off the doors
 - 6-2) Leave the shelves in place so that children may not easily climb inside
- 7) Unplug the freezer before carrying out user maintenance on it.
- 8) This freezer can be used by children age eight years and older and persons with reduced physical or mental capabilities or lack of experience and knowledge if they are given supervision or instruction concerning the use of the freezer in a safe way and understand the hazards involved. Children should not play with the freezer. Cleaning and maintenance should not be performed by children without supervision.
- 9) If a component part is damaged, it must be replaced by the manufacturer, its service agent, or similar qualified persons in order to avoid a hazard.
- 10) Please dispose of the freezer according to local regulations as the freezer contains flammable gas and refrigerant.
- 11) Follow local regulations regarding disposal of the freezer due to flammable refrigerant and gas. All refrigeration products contain refrigerants, which under the guidelines of federal law must be removed before disposal. It is the consumer's responsibility to comply with federal and local regulations when disposing of this product.
- 12) This freezer is intended to be used in household and similar environments.

13) Do not store or use gasoline or any flammable liquids inside or in the vicinity of this freezer.

14) Do not use extension cords or ungrounded (two-prong) adapters with this freezer. If the power cord is too short, have a qualified electrician install an outlet near the freezer. Use of an extension cord can negatively affect the freezer's performance.

Grounding requirement

This freezer must be grounded. This freezer is equipped with a cord having a grounding wire with a grounding plug. The plug must be inserted into an outlet that is properly installed and grounded.

Improper use of the grounding plug can result in a risk of electric shock. Consult a qualified electrician or service person if the grounding instructions are not completely understood, or if doubt exists as to whether the freezer is properly grounded.

2.2 Safety instruction for refrigerant



WARNING



Explosion Hazard.

Keep flammable materials and vapors, such as gasoline, away from freezer. Failure to do so can result in fire, explosion, or death.



Safety instruction for refrigerant

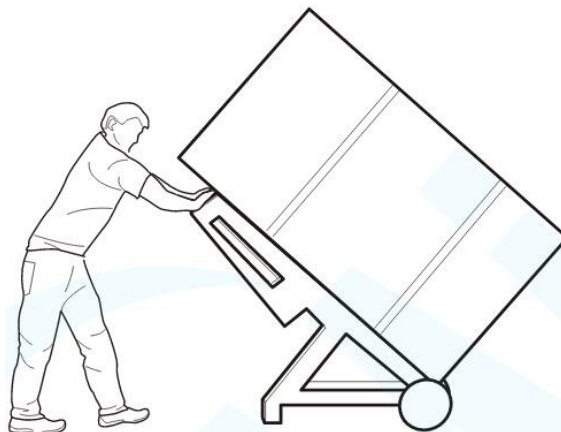
DANGER—Risk of Fire or Explosion. Flammable Refrigerant Used. To Be Repaired Only By Trained Service Personnel. Do Not Use Mechanical Devices. Do Not Puncture Refrigerant Tubing. CAUTION—Risk of Fire or Explosion. Flammable Refrigerant Used. Consult Repair Manual/Owner's Guide Before Attempting To Service This Product. All Safety Precautions Must be Followed. CAUTION—Risk of Fire or Explosion. Dispose of Properly In Accordance With Federal Or Local Regulations. Flammable Refrigerant Used. CAUTION—Risk of Fire or Explosion Due To Puncture Of Refrigerant Tubing; Follow Handling Instructions Carefully. Flammable Refrigerant Used.

3. Transport

3.1 Handling

Handling

- 1) Protect the refrigerator in moving it, Same as shown as left photo, please move it by handcart with cushion
- 2) Remove all packing materials and bottom cushion, the move into house for placement
- 3) After moving it to appropriate location, wait for 2 hours before power on.



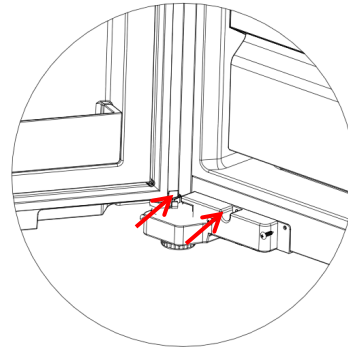
4. Installation and commissioning

4.1 Door Disassembly and Assembly

The refrigerator door needs to be dismantled if it cannot enter the room in the whole.

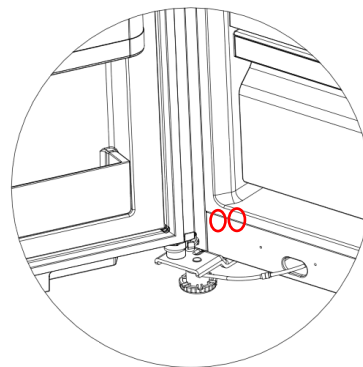
Disassembly of Freezer door

1) Use cross screwdriver to remove the screws by anticlockwise, and then take out the protective cover.

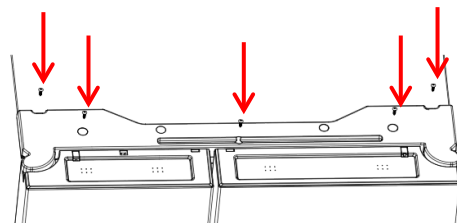


2) Pull out the spacing guard sheet at the two ends of water connector, and then push the locked cushion inwards, meantime pull out water pipe.

Note: please make sure the fast connector can't be left at the end of water pipe of door

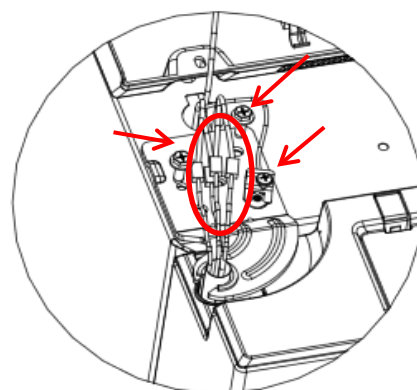


3) As shown as right photos, use screwdriver to remove the 5 pcs screws, and then remove the upper hinge cover.

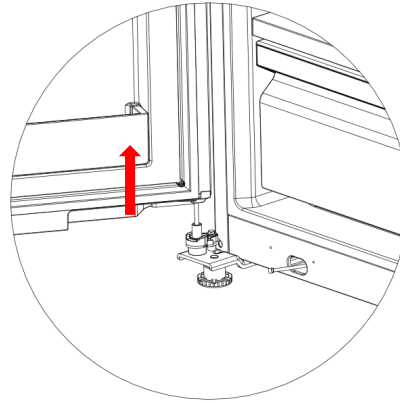


1) Disconnect the 5pcs fast connectors, then use cross screwdriver or socket spanner to remove the 2pcs M5 screws and 1 pc grounding screw by anticlockwise.

Note: please make sure the freezer door fit closely to the cabinet; otherwise the refrigerator door may fall down during above operation.



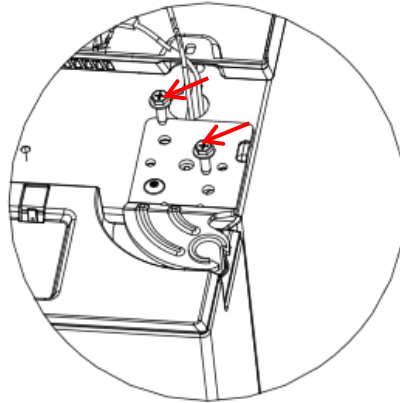
5) Lift up the freezer door until the hinge axis separated from the axis hole of door, then carry the door to a suitable place.



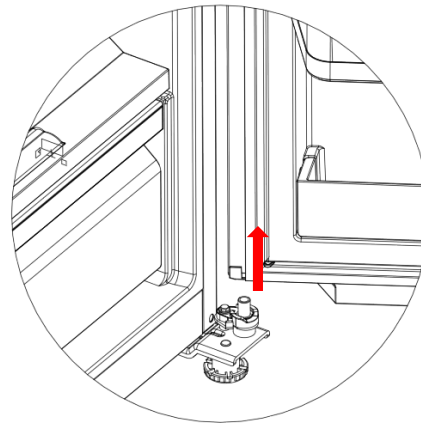
Disassembly of refrigerator door

1) Use cross screwdriver or socket spanner to remove the 2 pcs M5 screws by anticlockwise, and then remove the upper hinge.

Note: please make sure the refrigerator door fit closely to the cabinet; otherwise the refrigerator door may fall down during above operation.

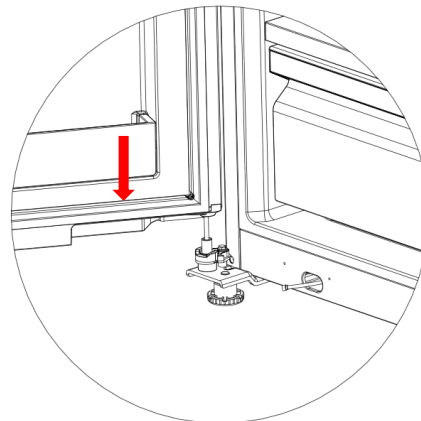


2) Lift up the refrigerator door until the hinge axis separated from the axis hole of door, then carry the door to a suitable place.

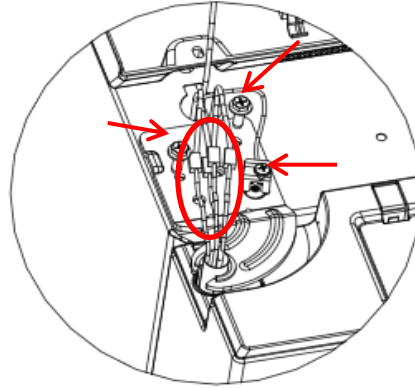


Assembly of freezer door

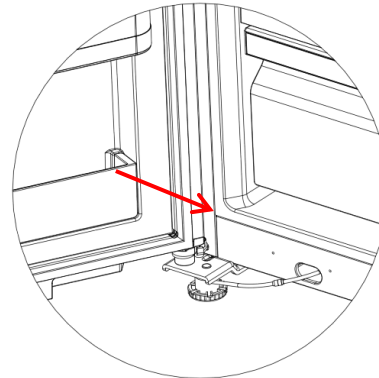
1) Put the freezer door down gently until the axis of hinge inserted into the axis hole of door completely.



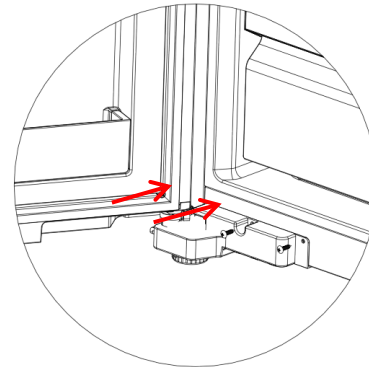
2) Put the upper hinge on suitable position, then use cross screwdriver or socket spanner to fix the 2 pcs M5 screws and 1 pc grounding screw by clockwise, and connect the 6 pcs fast connectors.



3) Push the locked cushion of water connector inwards, then insert the water pipe into the water connector to the end, and finally assemble the spacing guard sheet.
Note: the water pipe can't be connected incorrectly, only the water pipes in same color can be connected

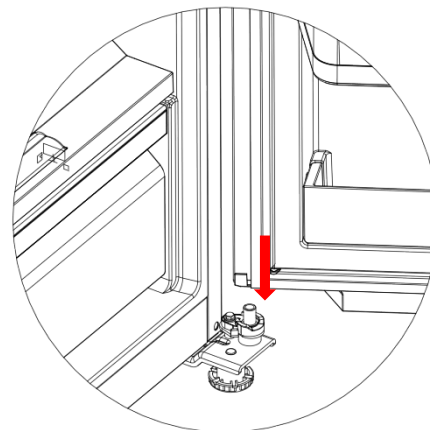


4) Assemble the protective cover and fix the screws

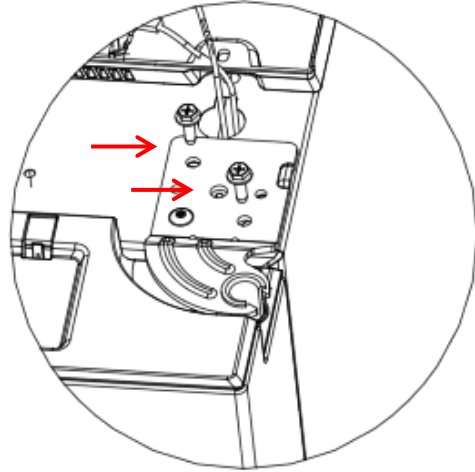


assembly of refrigerator door

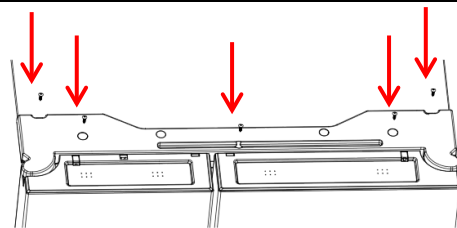
1) Put the refrigerator door down gently until the axis of hinge inserted into the axis hole of door completely, and then closes the door, to make sure the refrigerator door fit the cabinet closely.



2) Put the upper hinge on suitable position, then use cross screwdriver or socket spanner to fix the 2 pcs M5 screws.



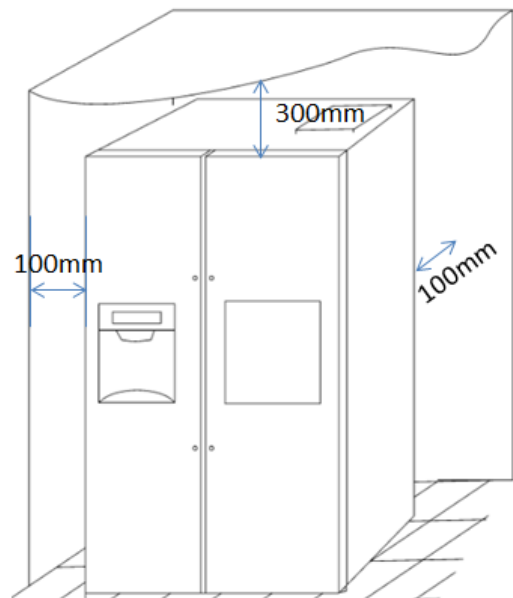
3) Put the hinge cover on suitable position, and then use the screwdriver to fix the 5 pcs screws.



4.2 Installation location

Installation location

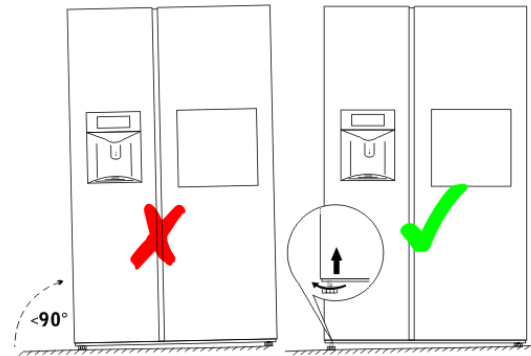
Location that is easy for ventilation shall be chosen to facilitate heat dissipation, enhance its performance and reduce the energy consumption.



4.3 Leveling of the refrigerator

Leveling of the refrigerator

If the refrigerator cannot be placed steadily, adjust the footing to level it.



4.4 Left or right open door reversal (None)

Door reversal

Door reversal	None
---------------	------

4.5 Installation of handle (None)

Installation of handle

Installation of handle	None
------------------------	------

4.6 Installation of door lock (None)

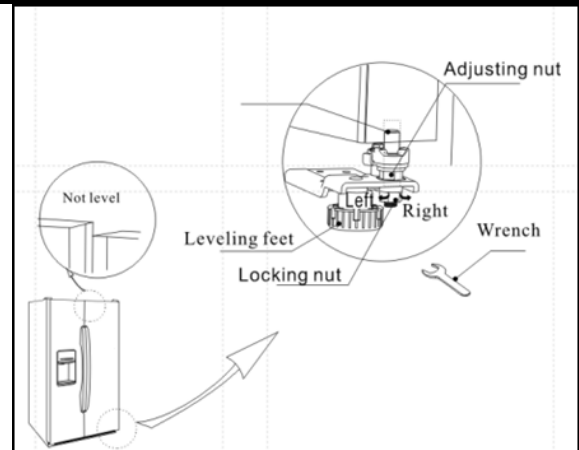
Installation of door lock

Installation of door lock	None
---------------------------	------

4.7 Adjustment to level the door

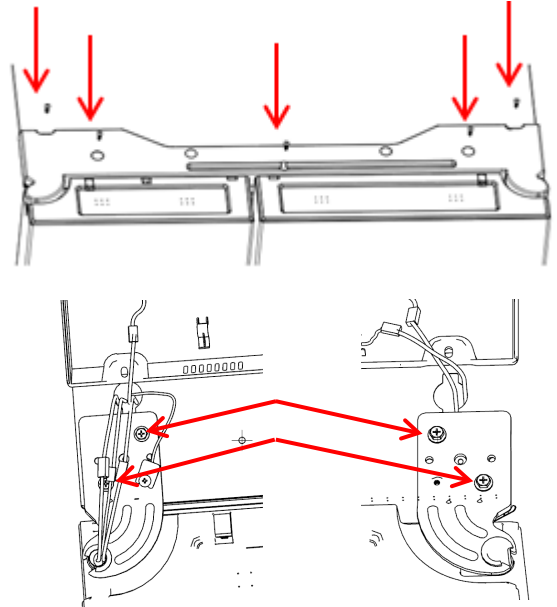
Adjustment to level the door (Up and down)

- 1) Refrigerator door lower: loosen locking nut counterclockwise, then rotate the adjusting nut counterclockwise to adjust the height of door, at last tighten the locking nut clockwise.
- 2) Refrigerator door higher: loosen locking nut counterclockwise firstly, then rotate the adjusting nut clockwise, at last tighten the locking nut clockwise.



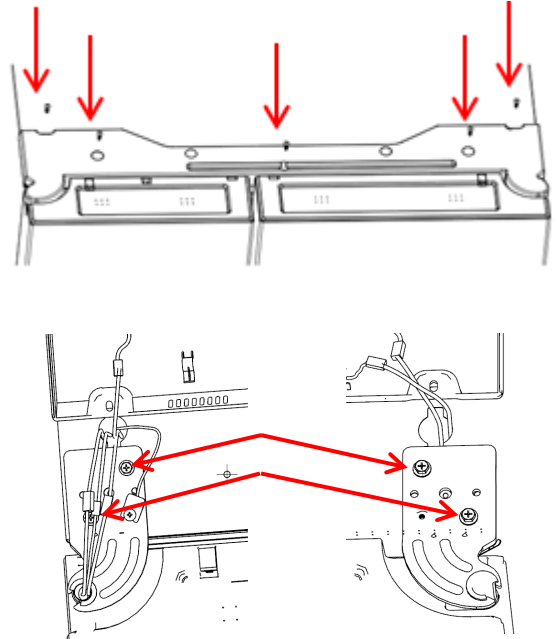
Adjustment to inhomogeneous aperture the door (Up and down)

- 1) Use screwdriver to remove the 5 pcs screws, and then remove the upper hinge cover.
- 2) Use screwdriver to loosen the 2 pcs screws, right and left adjust the upper hinge to make door aperture be homogeneity.
- 3) Put the hinge cover on suitable position, and then use the screwdriver to fix the 5 pcs screws.



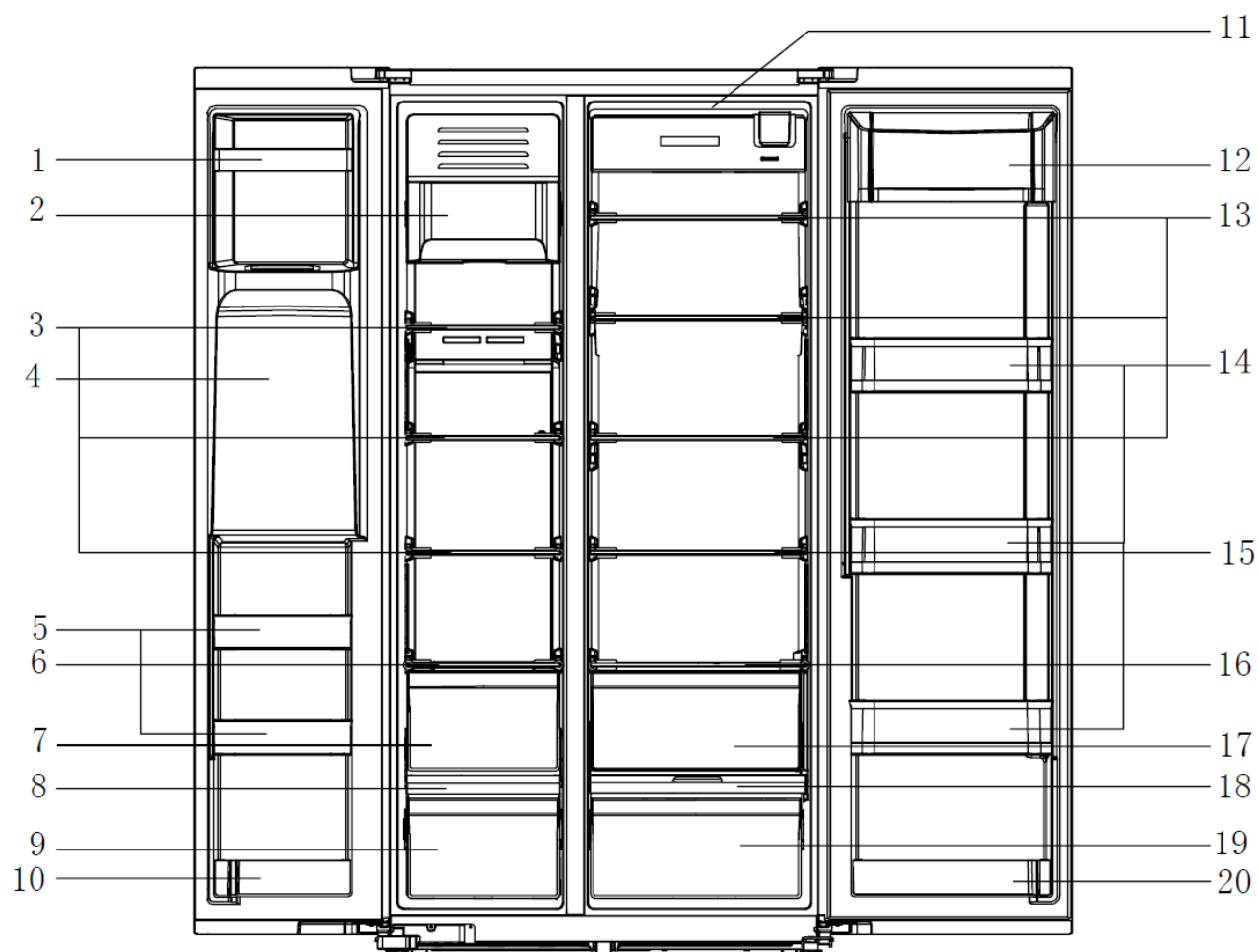
Adjustment to plane the door (left and right)

- 1) Use screwdriver to remove the 5 pcs screws, and then remove the upper hinge cover.
- 2) Use screwdriver to loosen the 2 pcs screws, front and back adjust the upper hinge to make door aperture be homogeneity.
- 3) Put the hinge cover on suitable position, and then use the screwdriver to fix the 5 pcs screws.



5. Product configuration and dimension

5.1 Main parts and their names

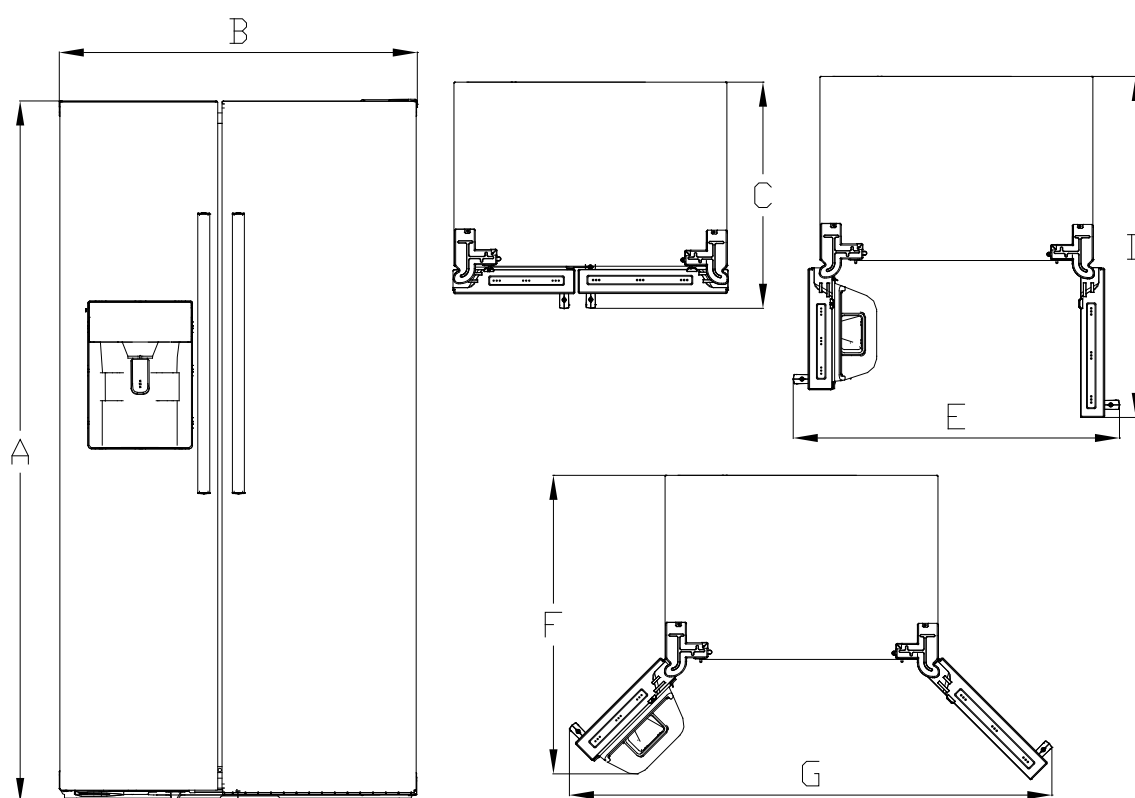


(The picture is only for reference, and specific appearance and configuration are subject to the real product)

Freezer chamber	Refrigerator chamber
1 door rack	11 LED lighting
2 ice tank	12 door rack
3 shelf	13 shelf
4 Dispenser	14 door rack
5 door rack	15 shelf
6 shelf	16 shelf
7 drawer	17 vegetable crisper
8 shelf	18 shelf
9 drawer	19 vegetable crisper
10 door rack	20 door rack

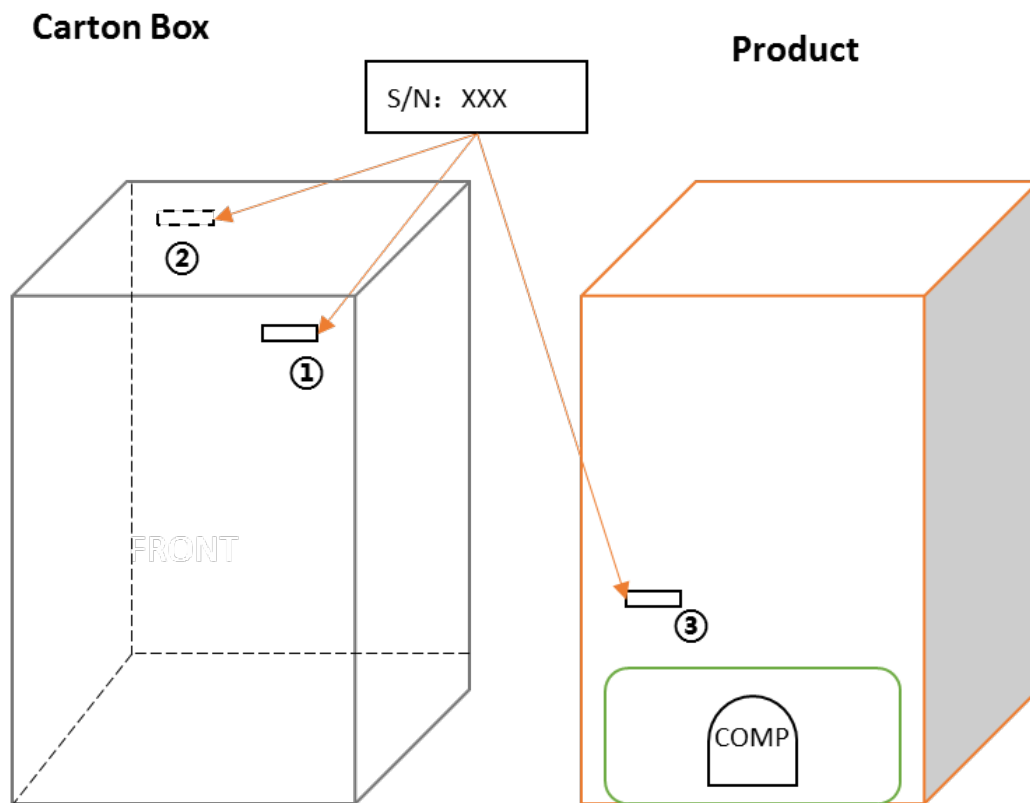
5.2 External dimension

Description	Code	Size (mm)
Height to Top of Case	A	1775
Width	B	911
Depth w/Handle	C	860
Depth w/Handle	C1	857
Depth (Door open 90 deg. w)	D	1273
Width (Door open 90 deg. w)	E	1039
Depth (Door open 135 deg. w)	F	1226
Width (Door open 135 deg. w)	G	1560



(The picture is only for reference)

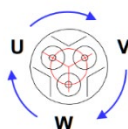
5.3 Location of S/N



Some products also have S/N on the lower part of the right side of the Cabinet.

6. Product specification

6.1 Electrical parameters

Applicable Models	HC-970WE	
Product Name	UR-BCD746WE-DT	
Product Code	22031050000741	
Item	Specification	Specification
Refrigerant	R600a	
Compressor	DZ120V1U	
Starter(PTC)	None	
Overload protector(OLP)	B110-105	
Integrate PTC+OLP	None	
Variable frequency driver board	HGFM/220-240V(50/60)Hz	
Capacitor	None	
Power filter	Have	
Power reactor	None	
Winding resistance of compressor wiring terminal	Rmc: 9.30Ω±5% Rsc: 9.30Ω±5% Rms;9.30Ω±5%	
Winding resistance picture		
The input power of compressor	165W	
Motor		
Fan motor of the freezing chamber	DC12V/≤3.5W	
Electric damperof the refrigerating chamber	DC12V 60mA	
Condensation fan	DC12V 2.5W	
separation the ice motor	DC12V 6W	
ice output motor	DC165V	
Open door motor	DC12V	
Lights inside the refrigerator		
Lights inside the freezing chamber	12V≤2.5W	
Lights inside the refrigerating chamber	12V≤2.5W	
othersLights	5V≤1W	
Switch of the refrigerator door	☐ Mechanical switch ☒ Magnetism control switch	

6.2 Cooling temperature setting range

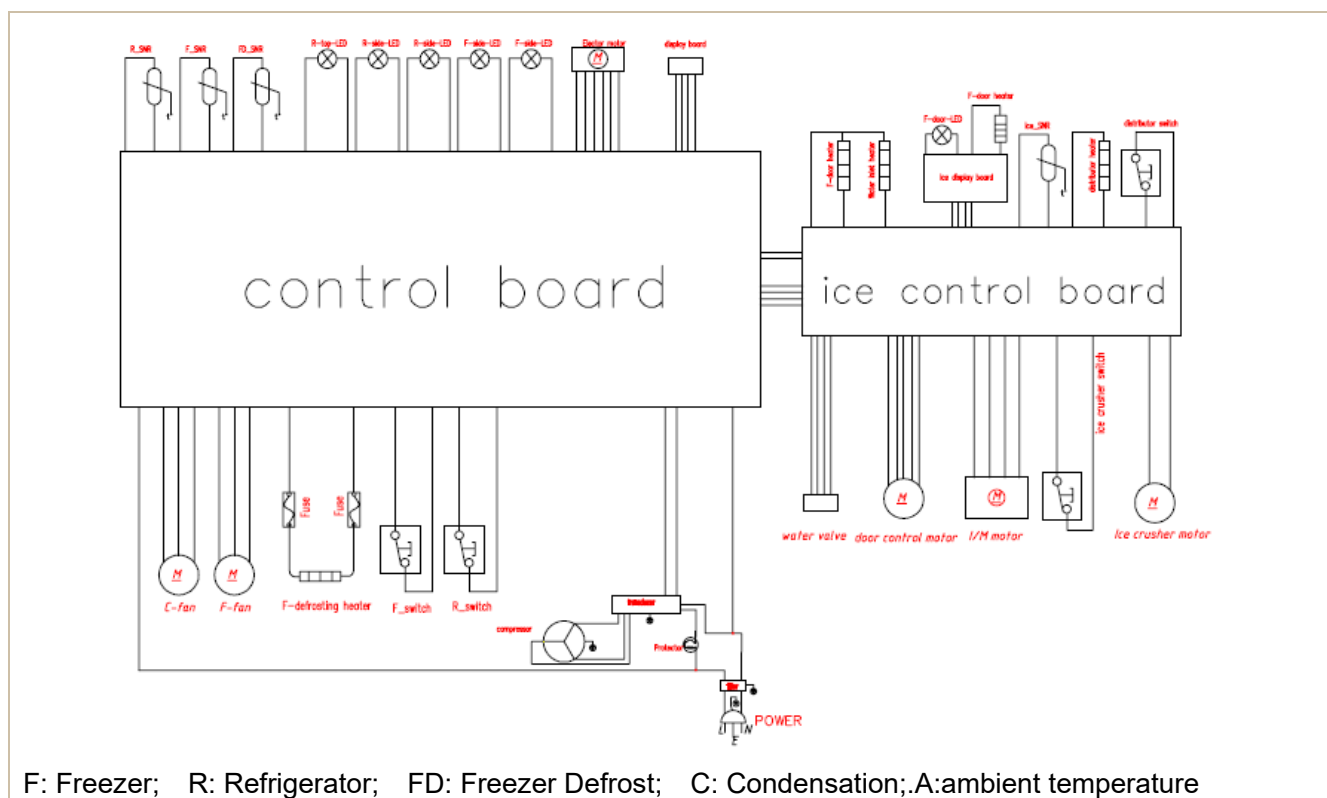
Compartment	The highest (°C)	Lowest (°C)
Freezing	-14	-22
Refrigerating	7	1

6.3 Defrosting parts

electronic type

Item	Specification
Defrosting sensor	NTC B3839
Fuse	77°C
Defrost heater in freezing chamber	115V 240W

6.4 Circuit diagram



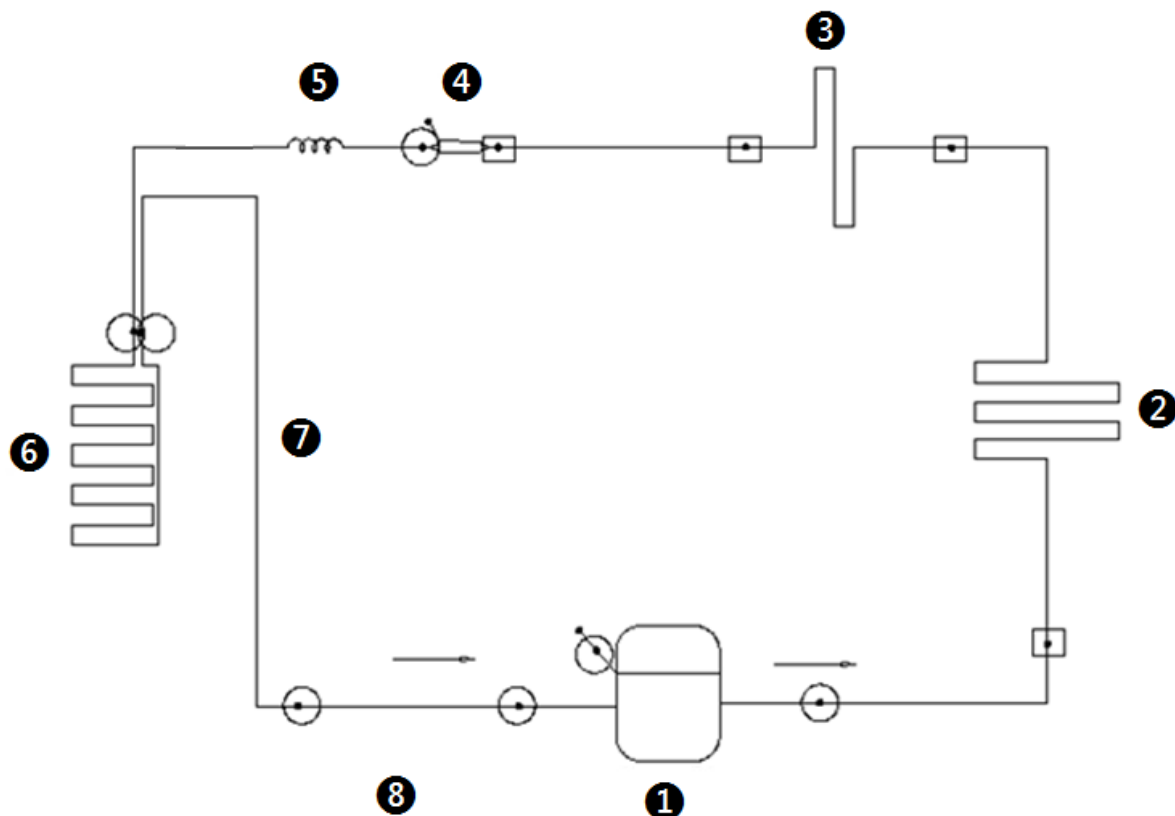
6.7Water valve

Model	Specification
/	115V 20W

7.Refrigerating piping system and circulating route of cooling air

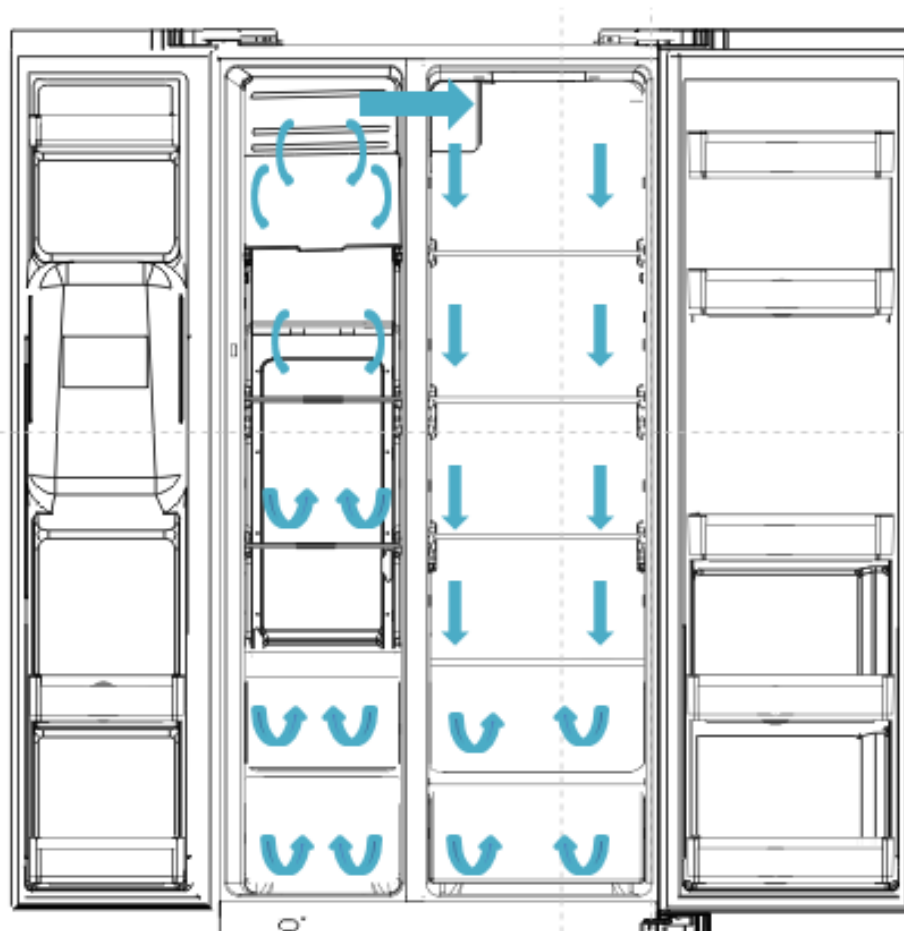
7.1 Refrigerating piping system

① Compressor→② Condenser→③ Anti-water condensed pipe→④ Drier→⑤ Capillary Tube→⑥ Evaporator→⑦ Suction Pipe→⑧ Suction connection Pipe→① Compressor



7.2Special soldering position (None)

7.3Circulating route of cooling air



8. Dismantling of parts

8.1 Parts on the door

Door seal

Door seal is installed into door liner groove.

- 1) Open the refrigerator door;
- 2) Take the door seal ① out of door liner;



Door tray

While squeezing it inward, lift up the baffle and take it out from refrigerator liner.



8.2 Parts inside the refrigerator

Shelves

- 1) Lift up the division plate with a proper force and pull it out towards yourself;



Freezer Drawer

The drawer is located at the freezing chambers;
 1) Pull the drawer out completely;
 2) Lift it up slightly and take it out from the refrigerator.



8.3 Light system

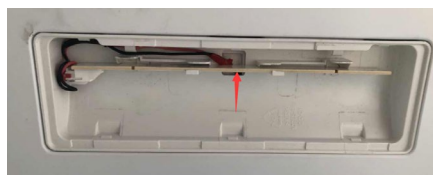
Light

Light of the refrigerating

1) Use vacuum cap to pull the lampshade outwards



2) Push away LED light panel with your hand along the arrow direction shown in the picture and separate LED light panel from the hook; then take down LED light panel



3) Remove the connecting harness terminals on LED light panel and take down the LED light panel.



Light of the freezing

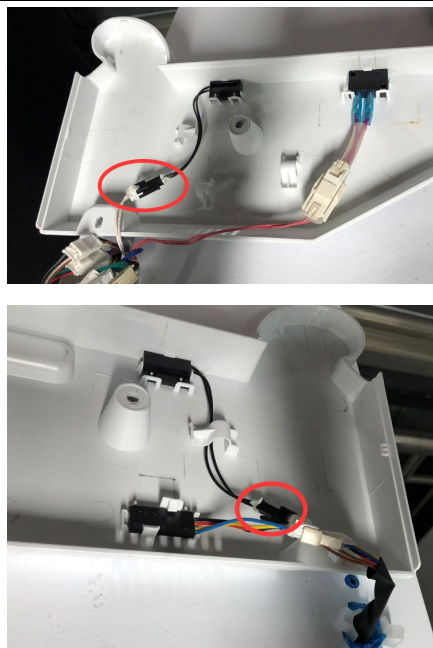
Refer to the method of disassembling the freezer light.

Light switch

1) Use screwdriver to remove the 5 pcs screws, and then remove the upper hinge cover.



2) Pull the terminal,remove the light switch.



Fresh light

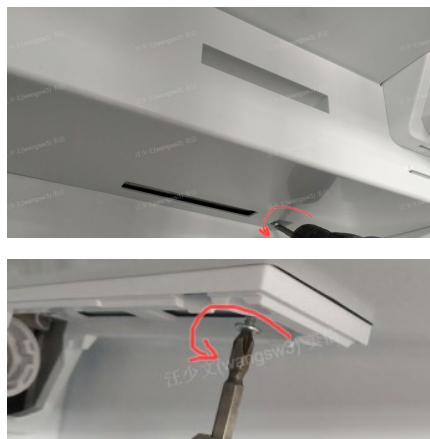
Fresh light

None

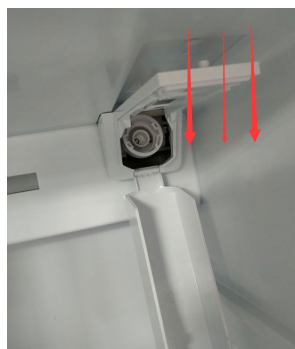
8.4 Air duct components refrigerating chamber and fan motor

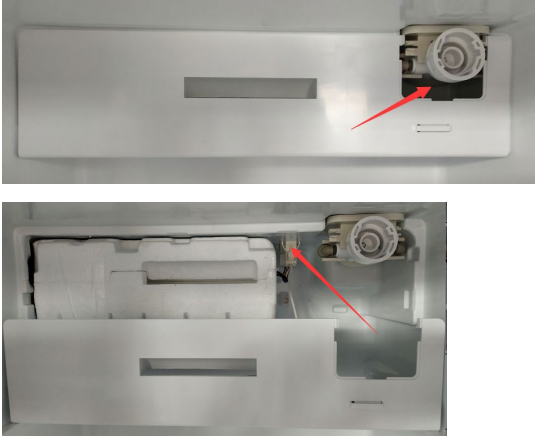
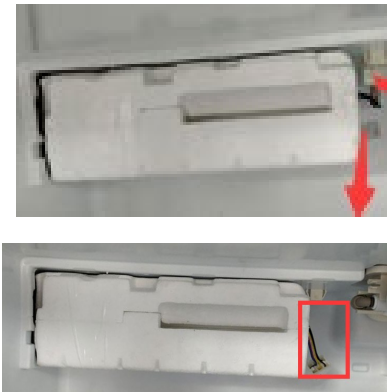
Air duct components refrigerating chamber

1) Remove by cross screwdriver the screws on the positions shown in the picture anticlockwise.



2) Remove the filter along the arrow direction, then remove the decoration cover of refrigerator air duct until air duct decoration cover completely falls off from refrigerator body and air duct foam, remove connecting terminals of sensors, then take out the decoration cover of refrigerating air duct.



	
3) Remove connecting terminals of ventilation door by hands, catch hold of air duct foam in refrigerating chamber and pull it towards the arrow direction shown in the picture until air duct foam is separated from refrigerator body, and take out refrigerating air duct foam.	
Fan motor of air duct	
	None
Electric damper	
	None

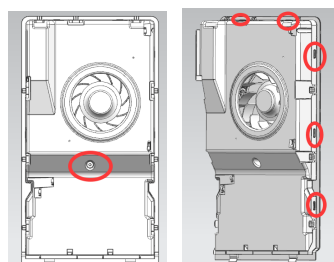
8.5 Air duct components in freezing chamber and fan motor

Air duct components in freezing chamber	
1) First, take out the screw cap of air duct cover with flat-blade screwdriver or blade; then, remove by cross screwdriver the two screws fixing air duct cover anticlockwise.	

2) As shown in the right picture, catch hold of upper part of freezing air duct as shown in the picture, pull it outward along the arrow direction with strength until air duct is separated from refrigerator body, remove the connecting harness between fan and refrigerator body, and take down freezing air duct.



3) Take out the one crew with cross screwdriver, pull the snap joint of upper air duct outward to make cover plate of upper air duct in freezing chamber fall off from air duct.

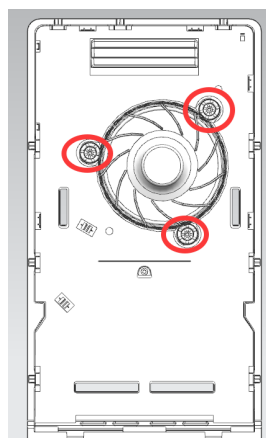


4) After the removal of upper air duct in freezing chamber. take out the one crew with cross screwdriver. Catch hold of cover plate of freezing lower air duct shown in the picture, pull it out along the arrow direction until it is separated from refrigerator body and take it out.



Fan motor of air duct

1) 冷冻室上风道盖板拆除后，用十字螺丝刀把三颗固定螺钉沿逆时针方向卸下。完成冷冻风扇电机的拆卸
After the removal of cover plate of upper air duct in freezing chamber. Remove the three fixing screws anticlockwise by cross screwdriver. Complete disassembly of the motor

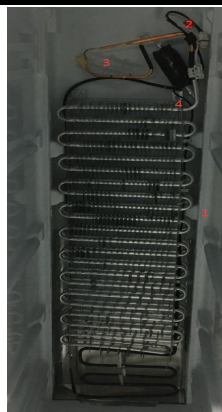


8.6 Evaporator and Defrost system

Evaporator in freezing chamber

Evaporator in freezing chamber

- 1) Remove the air duct components in freezing chamber.
- 2) Disconnect all connectors.
- 3) Remove the welding on inlet and outlet tubes.
- 4) Remove two screws which are used to fix the evaporator and remove the evaporator.



Components on the evaporator

Defrost thermostat

None

Defrosting fuse①

- 1) Disconnect the fuse connector.
 - 2) Cut off the band which fixes the fuse.
 - 3) Separate the fuse and the evaporator.
- *Don't break the welding of the evaporator in case that only the fuse needs to be replaced.



Defrost sensor②

The defrost sensor is located on top of the evaporator.

- 1) Disconnect the connector of defrost sensor
 - 2) Cut off the band which fixes the sensor.
 - 3) Separate the sensor and the evaporator.
- *Don't break the welding of the evaporator in case that only the sensor needs to be replaced.



Defrost heater③

The defrost heater is located at bottom of the evaporator.

- 1) Disconnect the connector of defrost heater.
 - 2) Cut off the band which fixes the defrost heater.
 - 3) Take off the defrost heater from the evaporator.
- *Don't break the welding of the evaporator in case that only the defrost heater needs to be replaced.



Evaporator in refrigerating chamber

None

Components on the evaporator

Defrosting fuse

None

Defrost sensor

None

Defrost heater

None

8.7 Compressor case

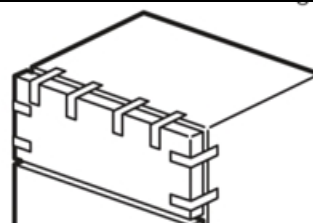
Rear cover

- 1) Remove by cross screwdriver the screws fixing back cover plate of compressor chamber anticlockwise
- 2) Take the back cover plate of compressor chamber upward.

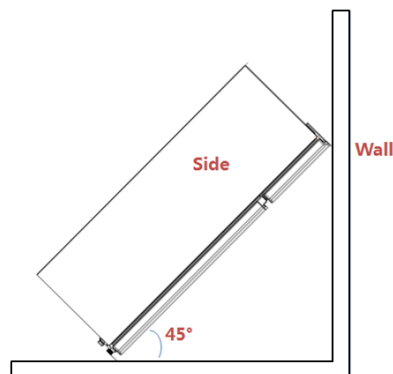


Compressor and the cooling system pipe

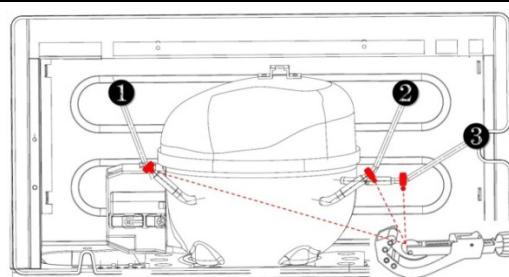
- 1) Cut off the power, remove the goods in the refrigerator, with the tape to make the door fixed firmly and prevent the door dropping when the refrigerator dumping.



- 2) Slowly tilt the refrigerator forward, relying on the wall or a solid enough object, leaving space to facilitate the operation. For safety, it should be carried by someone to prevent its falling.

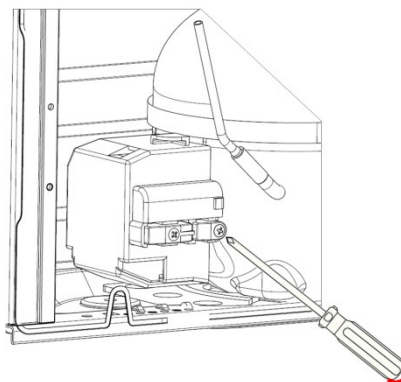


- 3) Cut off the compressor pipeline.
 - ① Cut off the process pipeline.
 - ② Cut off the low-pressure muffler.
 - ③ Cut off the high-pressure exhaust pipe.



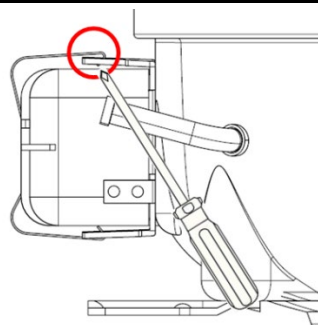
4-1) Remove the screws(for some models)

- Two screws outside
- One screw inside



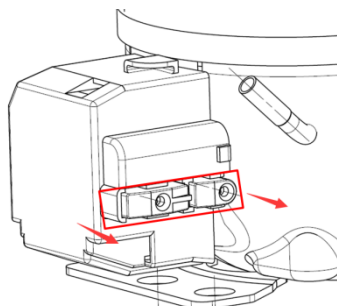
4-2) Remove the metal clamp(for some models)

- Disassembly the metal clamp that is fix the electric appliance shield



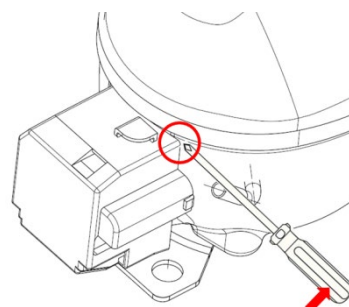
5) Remove the clipping strip

- Slowly pull it out



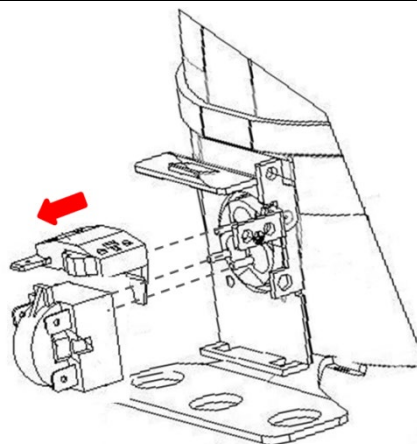
6) Remove the protective cover

- Pry the protective cover slowly from the upper part,
- Pull it out and remove it.

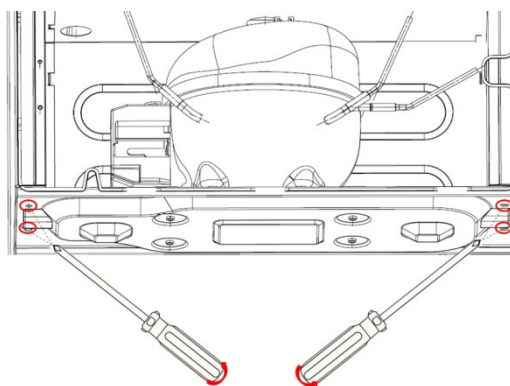


7) **Remove the starter and protector**

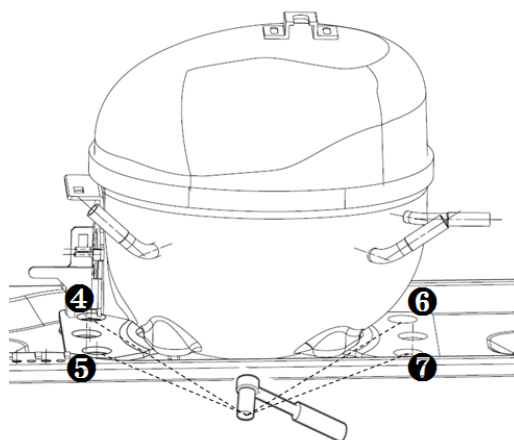
Unplug the starter and protector (you can use a screwdriver to pry it slowly)



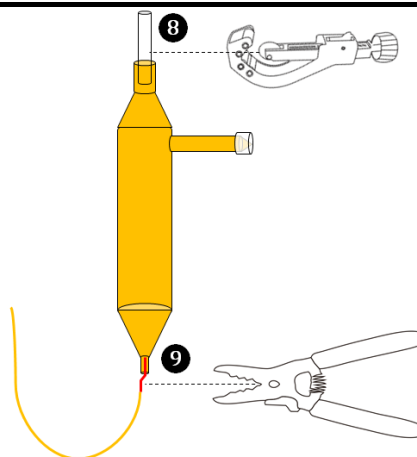
8) Loosen the screw of the compressor bottom plate, remove the floor together with the compressor from the box.



9) Use the wrench to remove the bolts by steps ④ ⑤ ⑥ ⑦, replace the compressor and reverse process can complete installation.

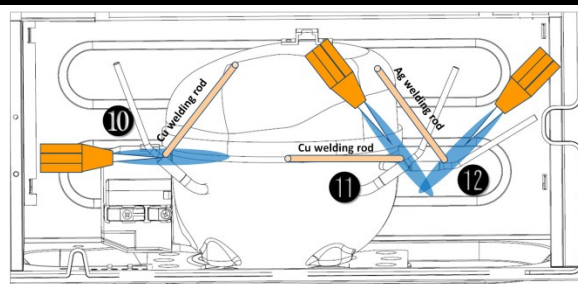


10) Use Pipe cutter cut off the condenser tube ⑧, then Shear off capillary ⑨ by the capillary tube scissors.

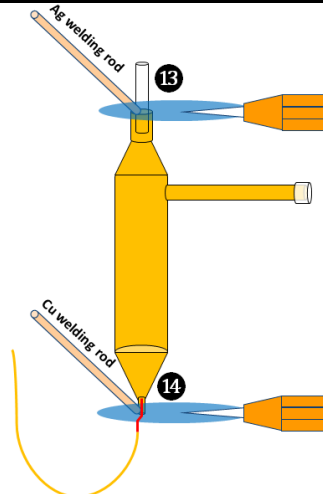


11) Replace the compressor and welding the compressor pipeline.

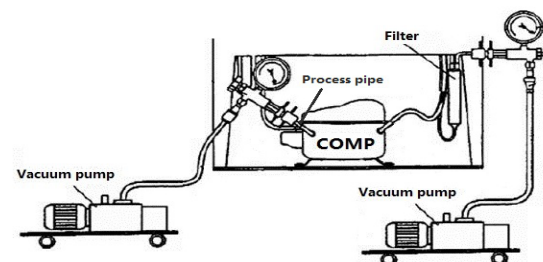
- ⑩ Welding the process pipeline.
- ⑪ Welding the low-pressure muffler.
- ⑫ Welding the high-pressure exhaust pipe.



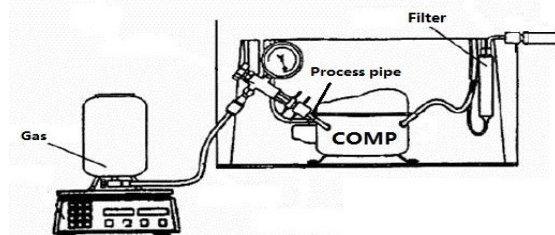
12) Replace the filter, Cu-Fe tubes welding ⑬ used Ag welding rod, Cu-Cu tubes welding ⑭ used Cu welding rod.



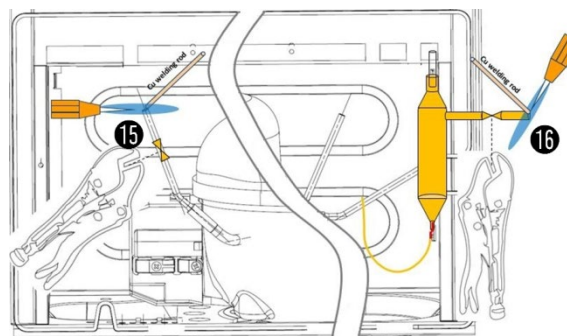
13) Vacuum system, The degree of vacuum below 6Pa.



14) Perfusion refrigerant.

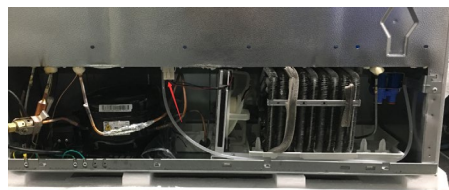


15) Use the vise grip pliers clamp the middle of the process pipe, then seal welding process tube ⑮ ⑯.

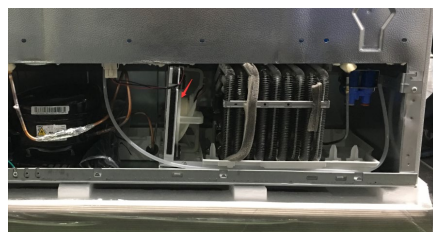


Condenser fan motor

1) Unplug the wiring connector by pressing the terminal lock



2) Loosen the screw, then pull out the fan bracket towards the direction what the arrows indicate.



Back hanging wire tube condensor

None

Piping system in the compressor case

① Suction Pipe

② Capillary Tube

③ Anti-water condensed pipe

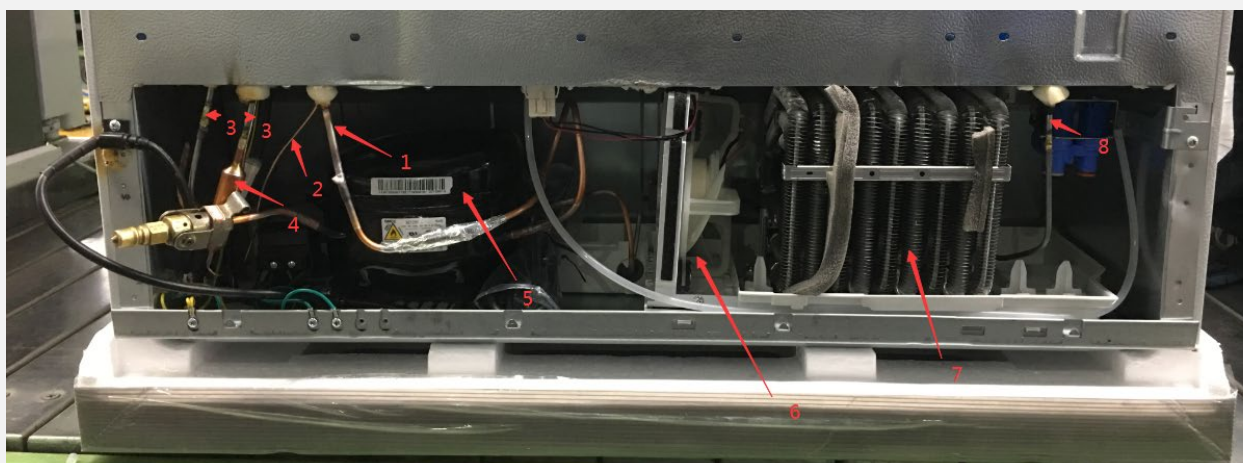
④ Drier

⑤ Compressor

⑥ Condenser fan

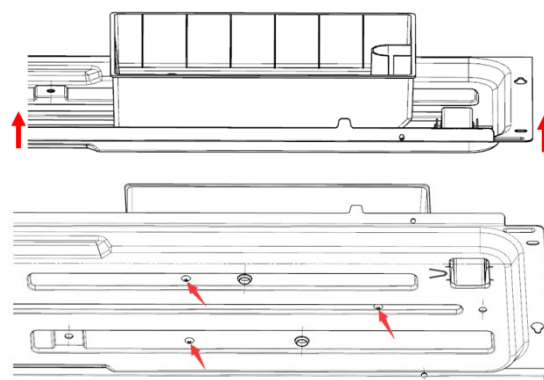
⑦ Condenser

⑧ Anti-water condensed pipe

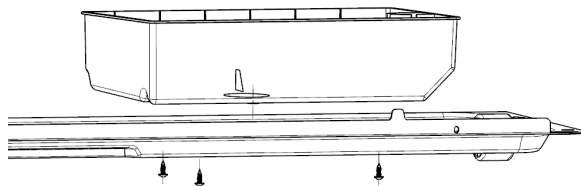


Drain tray

1) Remove the bottom screws of the compressor bottom plate



2) Replace the drain tray, the reverse process can complete installation.



8.8 Display control board

Display control board

1) Use vacuum cap to pull the control panel outwards



2) Disconnect the fast connector, then remove the control PCB



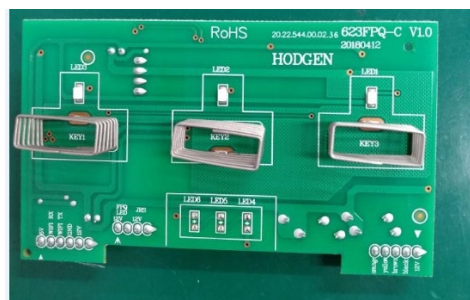
Dispenser display control panel

1) Use vacuum cap to pull the distributor display control panel outwards



2) Disconnect the fast connector, then remove the control PCB





8.9 Main control board

Have variable frequency driver board

1) remove the screw anticlockwise, then remove the PCB housing cover



2) Pull the lock of fast connector upwards, then disconnect the fast connector

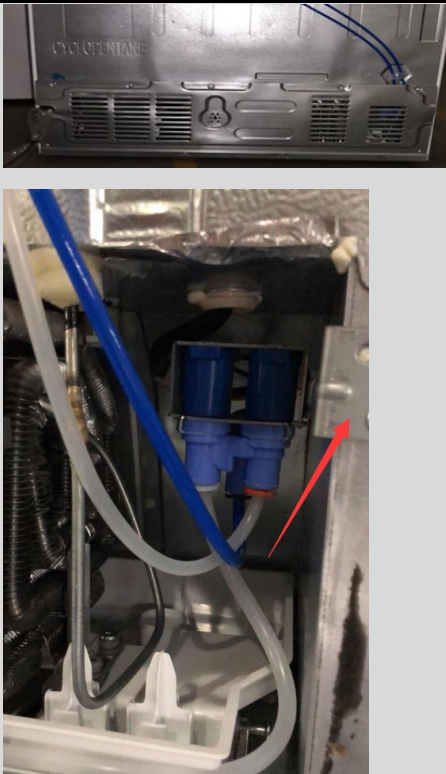


3) Remove the screw anticlockwise, then remove the PCB and variable frequency driver board.



8.10 Bar counter(None)

8.11 Water dispenser

Water dispenser	
Disassembly and installation of water valve	
1) Remove by cross screwdriver the screws fixing back cover plate of compressor chamber anticlockwise 2) Take the back cover plate of compressor chamber upward. 3) Unplug the three pipes 4) Remove the one screw fixing water valve, and take the water valve.	
Disassembly and installation of water tank	None

8.12Ice maker

disassembly of ice maker	
1) As shown in the photo, please up lift the ice storage box assembly and pull it out, then the ice storage box can be taken out.	

2) Pull the ice tray assembly outwards forcefully, and disconnect the wiring connector, then the ice tray assembly can be taken out from refrigerator.



3) Use cross screwdriver (anticlockwise) to remove the 4 screws of motor of ice releasing, and disconnect the wiring connector, then the motor can be taken out from refrigerator.

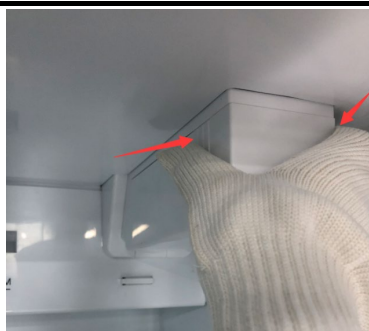


8.13 water filter component

Disassembly of filter(refer to A or B)

ADisassembly of filter built-in

1) Shut off the water supply, squeeze the mounting box of filter by hands, and detach the lid from the box.



2) Unplug the filter by rotating it counterclockwise



3) Fasten the mounting cover upward.



B Disassembly of filter external

None

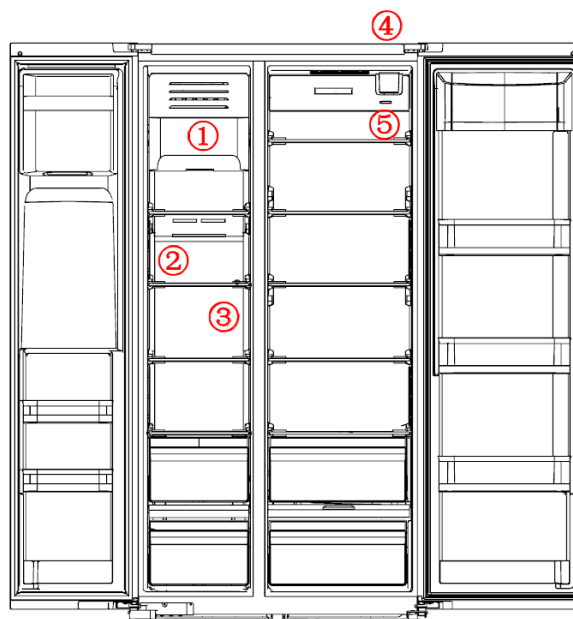
9.temperature sensing system

9.1 Position of sensors

Position of sensors

Have 5 sensors

- ① Ice machine sensor
- ② Sensor in freezing chamber
- ③ Defrost sensor freezing chamber
- ④ Ambient temperature or humidity sensor
- ⑤ Sensor in refrigerating chamber



9.2 Replacement of sensors

Sensor in freezing chamber

- 1) To remove the sensor cover, you may squeeze it up and down;
- 2) Take the sensor out from card slot;



Sensor in refrigerating chamber

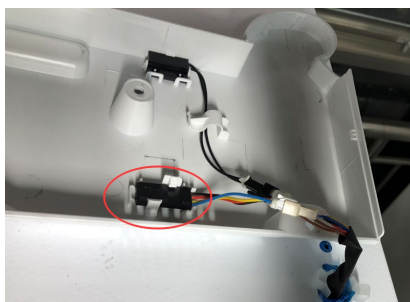
- 1) remove the decoration cover of refrigerator air duct until air duct decoration cover, refer to 1),2) of 8.4
- 2) Tear up the sponge, take the sensor out from card slot





Ambient temperature and humidity sensor

The sensor used for measurement of ambient temperature is located within upper hinge cover of refrigerating chamber door;



Defrost sensor in freezing chamber

The defrost sensor is located on top of the evaporator.

- 1) Disconnect the connector of defrost sensor
- 2) Cut off the band which fixes the sensor.
- 3) Separate the sensor and the evaporator.

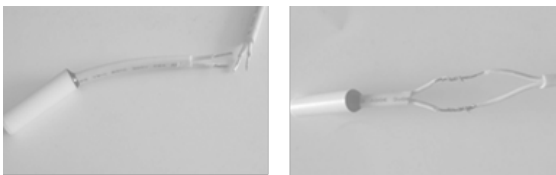
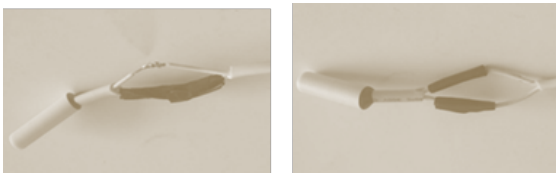
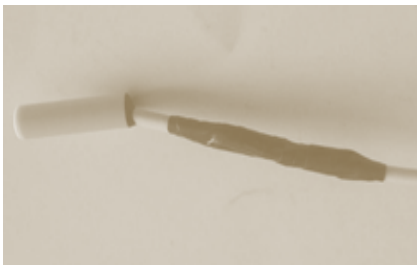


Ice machine sensor

The ice machine sensor is located on bottom of the ice-making box.



9.3 Sensor replacement

Sensor replacement guidelines①	
Cut off the damaged head of sensor.	
Strip off the sensor wiring.	
Take out a new sensor to cut the head of sensor. (Spare parts code: 11201007000795) Its technical specifications apply to all MIDEA refrigerators.	
Strip off the head of the sensor and connect it.	
Wrap the two wires together with insulation tape.	
Wrap the two wires together.	
Sensor replacement guidelines②	

Disconnect the fast connector



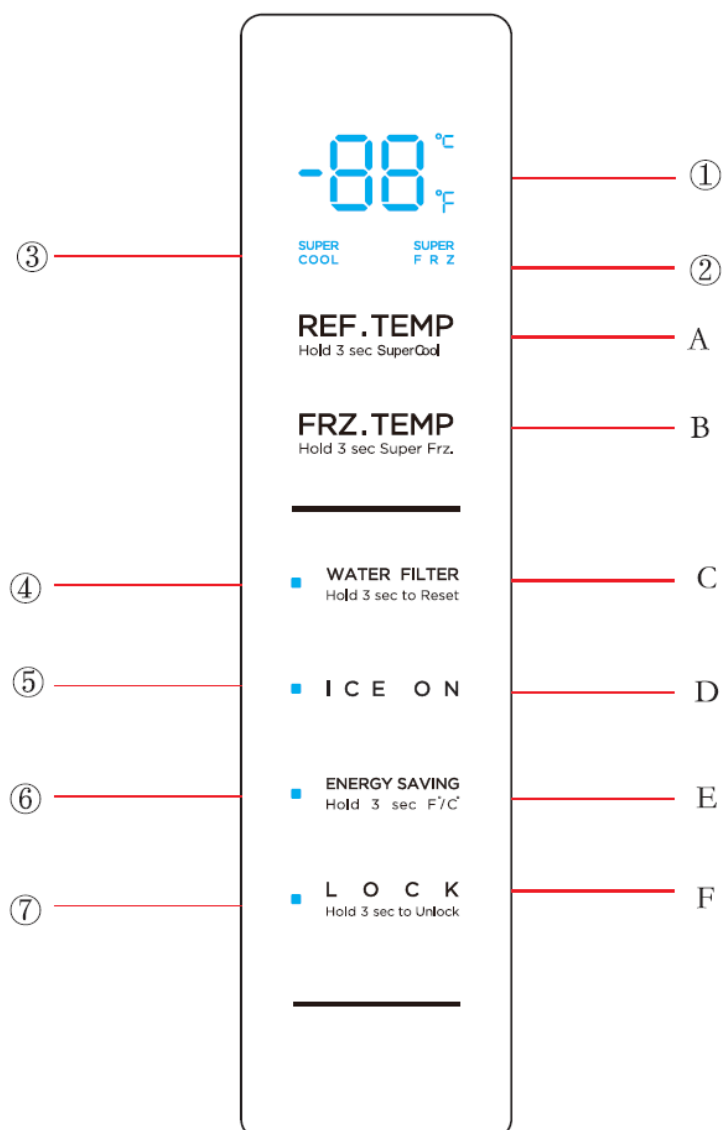
9.4 Sensor resistance(R/T)

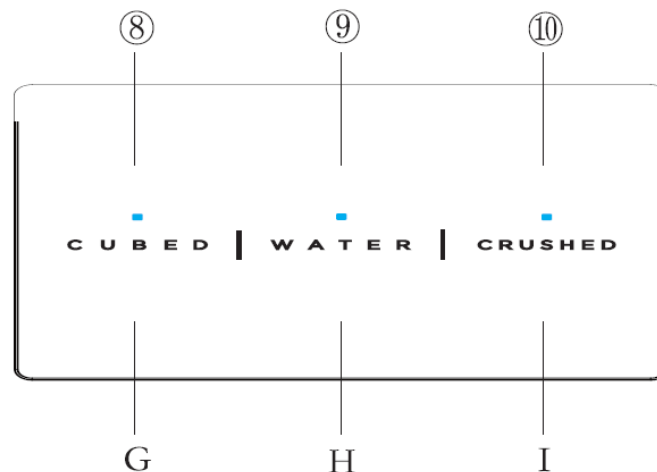
Tx(°C)	R (KΩ)	Tx(°C)	R (KΩ)	Tx(°C)	R (KΩ)	Tx(°C)	R (KΩ)	Tx(°C)	R (KΩ)
-30	33.81	-15	14.31	0	6.495	15	3.141	30	1.617
-29	31.85	-14	13.55	1	6.175	16	2.999	31	1.55
-28	30.01	-13	12.83	2	5.873	17	2.865	32	1.486
-27	28.29	-12	12.16	3	5.587	18	2.737	33	1.426
-26	26.68	-11	11.52	4	5.315	19	2.616	34	1.368
-25	25.17	-10	10.92	5	5.06	20	2.501	35	1.312
-24	23.76	-9	10.35	6	4.818	21	2.391	36	1.259
-23	22.43	-8	9.82	7	4.589	22	2.287	37	1.209
-22	21.18	-7	9.316	8	4.372	23	2.188	38	1.161
-21	20.01	-6	8.841	9	4.167	24	2.094	39	1.115
-20	18.9	-5	8.392	10	3.972	25	2.005	40	1.071
-19	17.87	-4	7.968	11	3.788	26	1.919	41	1.029
-18	16.9	-3	7.568	12	3.613	27	1.838	42	0.9885
-17	15.98	-2	7.19	13	3.447	28	1.761	43	0.9506
-16	15.12	-1	6.833	14	3.29	29	1.687	44	0.914

10. Function and operation

10.1 Operation panel

Icons	Button
① R/F temperature diagram	Ⓐ R temperature adjustment
② Quick freezing	Ⓑ F temperature adjustment
③ Quick refrigerating	Ⓒ Filter time reset button
④ Filter time reset icon	Ⓓ Ice machine on/off control button
⑤ Ice machine on/off icon	Ⓔ Energy saving mode button
⑥ Energy saving mode icon	Ⓕ Lock/unlock button
⑦ Lock/unlock icon	Ⓖ Whole ice mode
⑧ Whole ice mode icon	Ⓗ Water outlet mode
⑨ Water outlet mode icon	Ⓘ Crushed ice mode
⑩ Crushed ice mode icon	





10.2 Display

When the power is on for the first time, full display shall last for 3 seconds, the start-up tone shall sound at the same time and then enter into the normal operation display.

(When the power is on for the first time, the set temperature for the refrigerating chamber and freezing chamber is 39 Fahrenheit degree and 0 Fahrenheit degrees respectively, ice make opens by default and the default temperature of the first power on is Fahrenheit).

The fault code shall be displayed when there is a fault (cyclic display); the set temperature of the refrigerating chamber shall be displayed when there is no fault.

10.3 Temperature control

1) Temperature adjustment of refrigerator compartment

Press "REF. TEMP" button, then press "REF. TEMP" to adjust setting temperature, the setting temperature is changed circularly "7°C □ 6°C ... □ 1°C □ 7°C" or "44°F □ 43°F ... □ 34°F □ 44°F"

2) Temperature adjustment of Freezer compartment

Press "FRZ. TEMP" button, and then press "FRZ. TEMP" to adjust setting temperature, the setting temperature is changed circularly "-14°C → -15°C ... → -22°C → -14°C" or "7°F → 6°F ... → -7°F → 7°F"

3) Frequently-used MODE

Press "MODE" button, consumer can set "vacation" → "fast freezing" → "fast freezing + fast cooling" → "fast cooling" → "vacation" → "." circularly, the icons of corresponding modes will light on/off synchronously
Vacation mode: the freezer compartment will work as the setting temperature is -18°C, the refrigerator compartment is turn off;

Fast freezing mode: the freezer compartment will work as the setting temperature is -24°C, and after 26 hours the freezer compartment will quit this mode automatically and work according to last temperature setting;

Fast cooling mode: the refrigerator compartment will work as the setting temperature is 2°C, and after 150 minutes the refrigerator compartment quit this mode automatically and work according to last temperature setting;

When press "REF. TEMP" or "FRZ. TMEP" button, the "Vacation Mode" will be stopped, the refrigerator compartment and freezer compartment will work according to last temperature setting;

When press "FRZ. TEMP" button, the "Fast Freezing Mode" will be stopped, the freezer compartment will work according to last temperature setting;

When press "REF. TEMP" button, the "Fast Cooling Mode" will be stopped, the refrigerator compartment

will work according to last temperature setting;

10.4 Open door alarm

When refrigerator door or freezer door is open, the display panel is light on. There is notification tone when open door; if refrigerator door or freezer door is open last for 120s, there will be buzzer alarm, until the door is closed. Press any buttons on control panel can cancel this buzzer alarm.

Note: When open the door, the display panel is light on; when the door is closed, the display panel will be light off after 30s if there is no any operation on display panel

10.5 Function Selection

1) Mandatory mode (mandatory compressor starting, mandatory defrosting and mandatory ice making)

All below functions are only for diagnosis purpose, we advise to restart the refrigerator by power on/off if have used these functions.

Buttons operation:

Press both "LOCK/UNLOCK" and "FRZ.TEMP" button last for 3s to enter into/quit mandatory mode, the setting will take effect after locking.

Press "FRZ.TEMP" button, the ref. temperature zone will display "1" "2" "3" "4" circularly,

Different figures represent different functions:

- 1) When display "0", the refrigerator will not enter into anyone mandatory mode, and quit the mandatory mode.
- 2) When display "1", the compressor will start working last for 36 hours, then it will quit this mode automatically and back to normal working status.
- 3) When display "3", the ice maker will turn over the ice for twice, then intake water.

10.6 Control of ice maker

1) Ice maker on/off

Press "ICE ON" button to turn off/on ice maker, the default setting is **on**.

2) Options for ice /cooling water

Press the button of "CUBED"/"WATER"/"CRUSHED", the corresponding icons will light on. The working time of the power-on filter accumulates. When the filter works for 150 days, the filter icon shall light up to prompt the user to replace the filter and shall flash to prompt the user to replace the filter after 180 days. Long press the "filter time reset" button for 3 seconds to perform the function of "FILTER RESET". At this time the filter timing shall be reset;

10.7 Celsius Fahrenheit conversion

Long press "Energy Saving" button for 3s to convert the temperature from Celsius to Fahrenheit and Back again.

10.8 Fault code and solutions

Note: midea full common fault code, combined with the actual product display reference.

Fault code	Fault content	Steps for maintenance methods
E0	fault of ice maker	Step 1:Check whether the terminal of ice maker in main control board is well stuck, pull out the terminal and re-stick it

		in place Step 2: Check to see if there're foreign matters on the terminal Step 3: Check whether the terminal of ice maker is well
E1	Temperature sensor fault in refrigerating chamber	Step 1: Check whether the terminal of temperature sensor in main control board is well stuck, pull out the terminal and re-stick it in place Step 2: Check to see if there're foreign matters on the terminal Step 3: Inspect the refrigerating sensor whether contact is bad, and resend contact the fast connector Step 4: Replace main control board
E2	Temperature sensor fault in freezing chamber	Step 1: Check whether the terminal of temperature sensor in main control board is well stuck, pull out the terminal and re-stick it in place Step 2: Check to see if there're foreign matters on the terminal. Pull out the defrost sensor in freezing chamber according to the method in described in Article 9.2 and then inspect the sensor against the resistance value table in 9.3. Step 3: Replace main control board Step 4: Replace electrical wiring main harness
E3	Temperature sensor fault in variable chamber	Step 1: Check whether the terminal of temperature sensor in main control board is well stuck, pull out the terminal and re-stick it in place Step 2: Check to see if there're foreign matters on the terminal. Pull out the defrost sensor in variable chamber according to the method in described in Article 9.2 and then inspect the sensor against the resistance value table in 9.3. Step 3: Replace main control board Step 4: Replace electrical wiring main harness
E4	Defrost sensor fault in refrigerating chamber	Step 1: Check whether the terminal of defrost sensor in main control board is well stuck, pull out the terminal and re-stick it in place Step 2: Check to see if there're foreign matters on the terminal. Pull out the defrost sensor in refrigerating chamber according to the method in described in Article 8.6 and then inspect the sensor against the resistance value table in 9.3 Step 3: Replace main control board Step 4: Replace electrical wiring main harness
E5	Defrost sensor fault in freezing chamber	Step 1: Check whether the terminal of defrost sensor in main control board is well stuck, pull out the terminal and re-stick it in place. Step 2: Check to see if there're foreign matters on the terminal. Step 3: Inspect the defrost sensor whether contact is bad, and resend contact the fast connector

		Step 4: Replace main control board
E6	Communication failure	Step 1: Check whether the terminal CN9 is well stuck, pull out the terminal and re-stick it in place Step 2: Check to see if there're foreign matters on the terminal. Step 3: Inspect the display control board terminal whether contact is bad, and resend contact the fast connector Step 4: Replace main control board Step 5: Replace display control board Step 6: Replace main wire
E7	Ambient temperature sensor fault	Step 1: Check whether the terminal of temperature sensor in main control board is well stuck, pull out the terminal and re-stick it in place Step 2: Check whether the sensor wiring harness are connected. Step 3: Replace main control board. Step 4: Remove the temperature sensor at the hinge, and replace the temperature sensor.
E8	Defrost sensor fault in variable chamber	Step 1: Check whether the terminal of defrost sensor in main control board is well stuck, pull out the terminal and re-stick it in place Step 2: Check to see if there're foreign matters on the terminal. Pull out the defrost sensor in refrigerating chamber according to the method in described in Article 8.6 and then inspect the sensor against the resistance value table in 9.3 Step 3: Replace main control board Step 4: Replace electrical wiring main harness
E9	High temperature alarm in freezing chamber	Step 1: Check whether the door is tight shut. Step 2: Check whether the power has been off for a long time. Step 3: Check whether the compressor is not refrigerated. Step 4: Same to E2 maintenance methods.
EE	Circuit fault of ice maker sensor	Step 1: Check whether the terminal ice maker sensor of main control board is well stuck, pull out the terminal and re-stick it in place and check if there is any foreign matter on the terminal Step 2: Check whether the terminal of ice maker is well
EH	Circuit fault of ambient humidity sensor	Step 1: Check whether the terminal ambient humidity sensor of main control board is well stuck, pull out the terminal and re-stick it in place and check if there is any foreign matter on the terminal Step 2: Check whether the sensor wiring harness are connected. Step 3: Replace main control board. Step 4: Remove the humidity sensor, and replace the sensor.

EF	Circuit fault of water tank is not installed right in place	to check relevant connectors、whether the tank-base is installed rightly
CA	Motherboard ice maker communication failure	Check whether the connection of the terminal CN9 of main control panel and the terminal CN6 of ice maker plate is well stuck.
EP	Ice making machine off-ice fault	Step 1:Check whether the terminal of ice maker in main control board is well stuck, pull out the terminal and re-stick it in place Step 2: Check to see if there're foreign matters on the terminal Step 3: Check whether the terminal of ice maker is well
Cubed/water/crushed blink at the same time	Distributor control and ice maker control communication failure	Check whether the connection of the terminal CN9 of ice maker plate of and the terminal CN6 of distributor control is well stuck.

10.9 Defrosting function

Defrosting theory:

The defrosting of evaporator is realized by the heating of heater, following the temperature rise, the frost on evaporator becomes water, and the water flow into the evaporating pan via the draining system, the water in evaporating pan evaporate away finally

Defrosting steps:

Compressor switch off- -air duct closed---the fan in freezer chamber works for 3~10 min(for different kinds of refrigerators, the time is different too)---the heater start working---the heater switch off when the temperature rise to setting---defrosting done.

10.10 EYE ice machine Test mode

Enter Mode

Keep pressing the “FRE” and “Energy Saving” button for 3s, enter/exit EYE.

In Mode

Enter EYE, the display digital of ice machine mode will blink, and water gets into ice machine after

turning over the ice twice.

Exit Mode

1. Keep pressing the “FRE” and “Energy Saving” button for 3s, exit mode.
2. EYE mode is completed, exit mode.

10.11 Backup data for power fail

- 1) The running state of the refrigerator is remembered after compressor running for 1 hour continuously.
 - 2) The running state of the refrigerator is remembered after change function settings and lock.
 - 3) When the refrigerator is out of power and recharged, the running state of the refrigerator is same as before.
-

11. Compressor

11.1 Compressor on and off Control specifications

1.1 When one of the following conditions is met, the compressor stops:

- 1) $T_r \leq T_{rt}$;
- 2) The compressor runs continuously for more than 3 hours (Stop 5 minutes);

1.2 When all the following conditions are met, the compressor starts up:

- 1) $T_r \geq T_{rk}$;
- 2) Compressor downtime is more than 5min.

★When 1.1 and 1.2 are not satisfied, the compressor maintains the original state

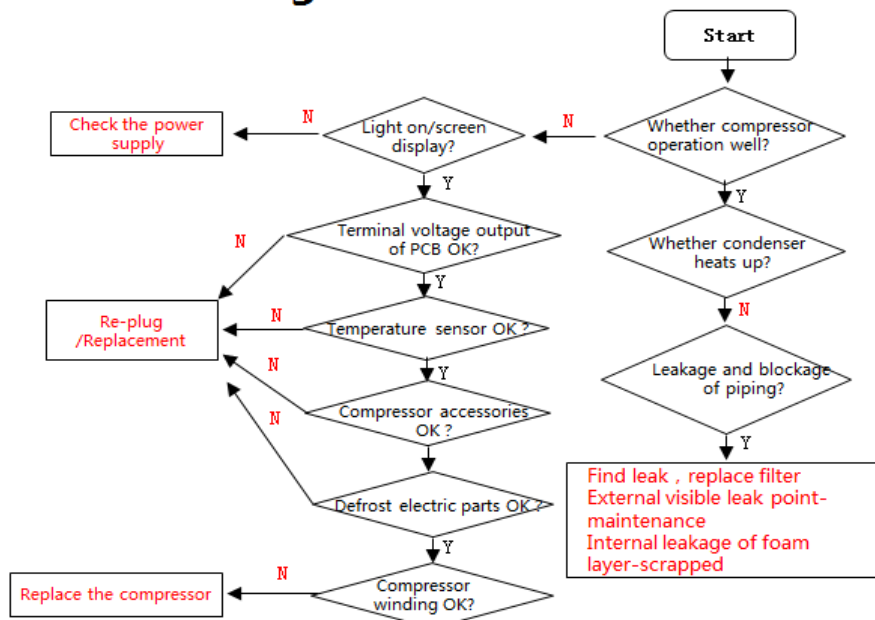
11.2 Variable frequency driver board fault analysis

running status of LED	Fault cause
not Light	not Working
Blink once: light 1S, extinguish1S	Standby
Blink once: light 0.5S, extinguish1.5S	Overcurrent
Blink twice: light 0.5S, extinguish0.5S, interval time(extinguish) is 1.5 second	Overvoltage
Blink four times: light 0.5S, extinguish0.5S, interval time(extinguish) is 1.5 second	Undervoltage
Blink eight times: light 0.5S, extinguish0.5S, interval time(extinguish) is 1.5 second	LOCK
Blink sixteen times: light 0.5S, extinguish0.5S, interval time(extinguish) is 1.5 second	Overload protection

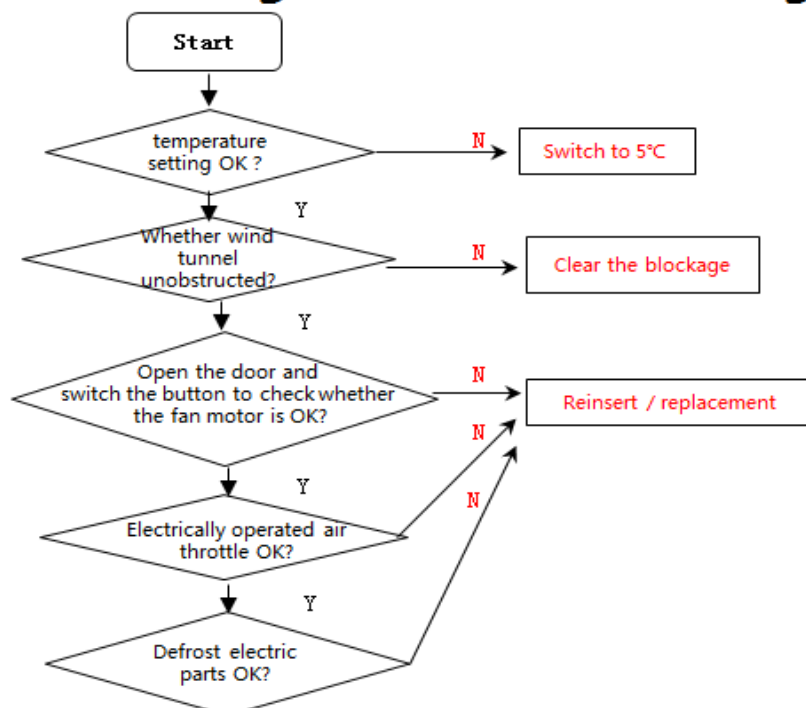
12. Troubleshooting Method

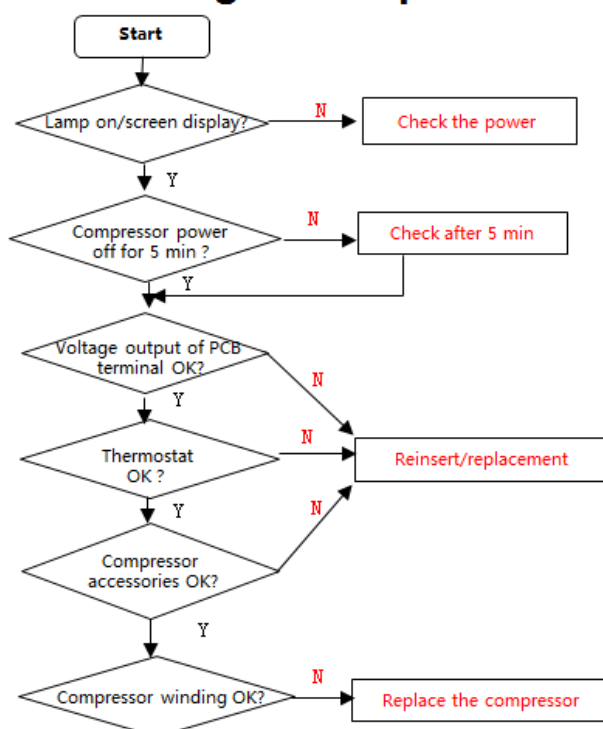
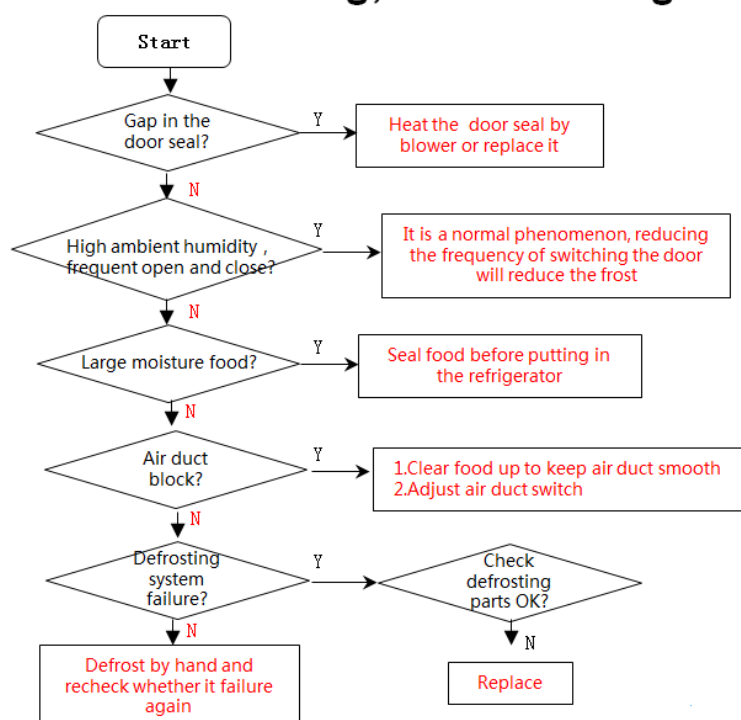
12.1 No cooling(Air cooling-Electronic)

No cooling of F room and R room

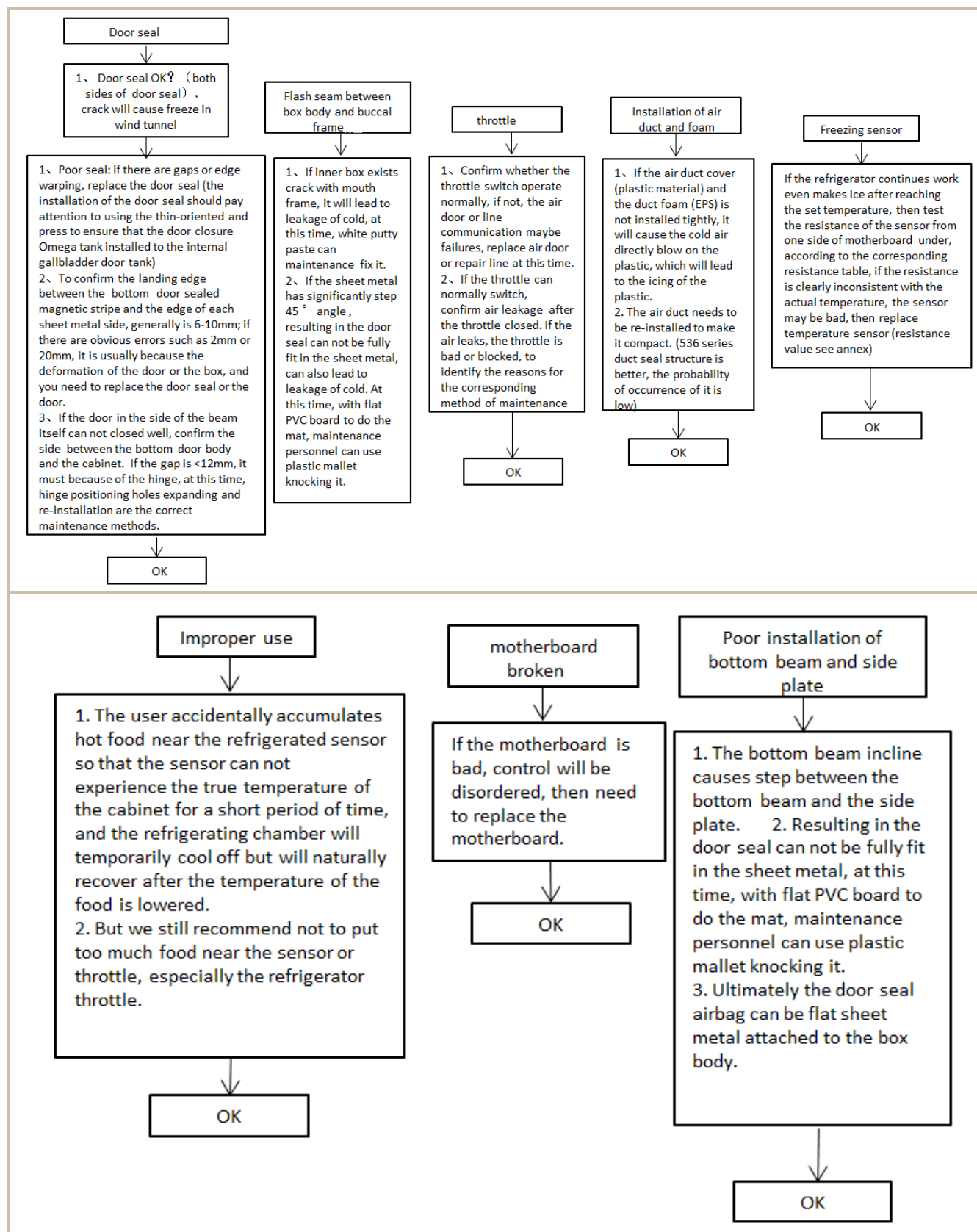


F room cooling, R room no cooling

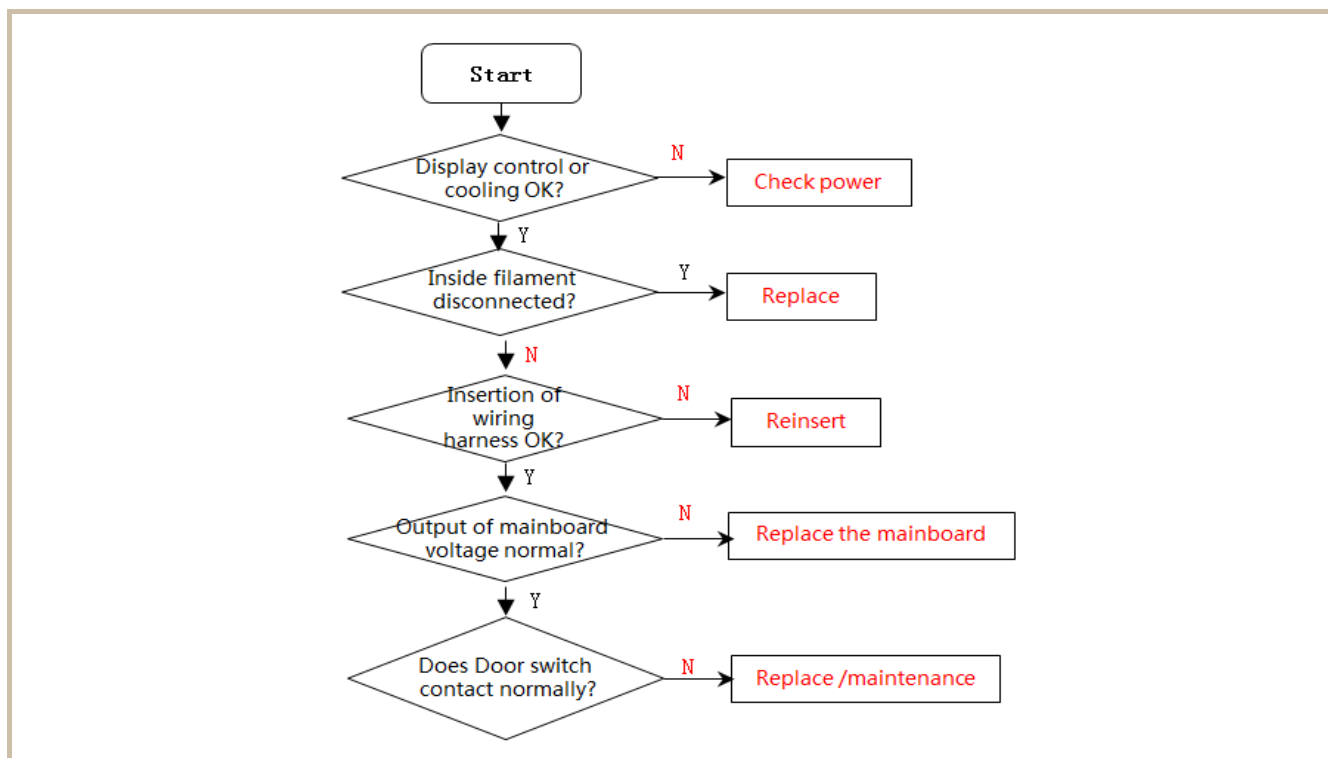


12.2 No working of compressor**No working of compressor****12.3 Inside frosting, no defrosting****Inside frosting, no defrosting**

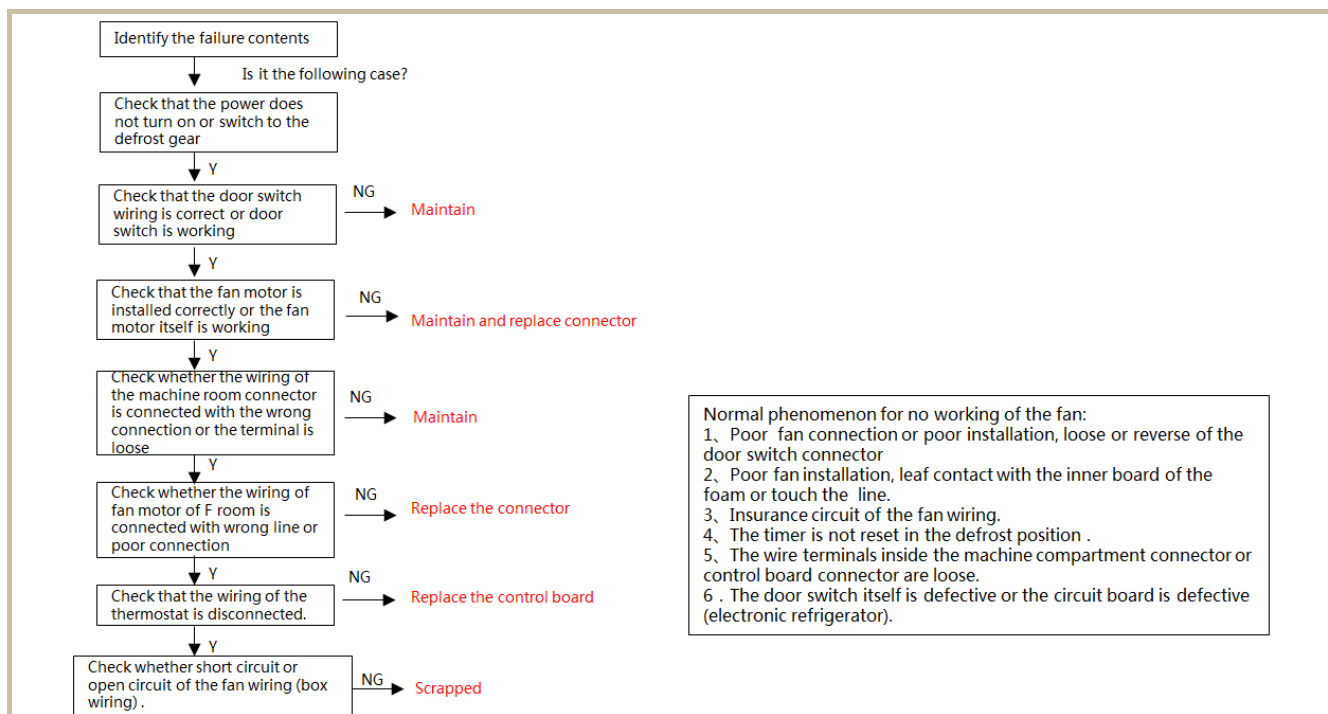
12.4 Inside frosting, no defrosting-Maintenance guidelines



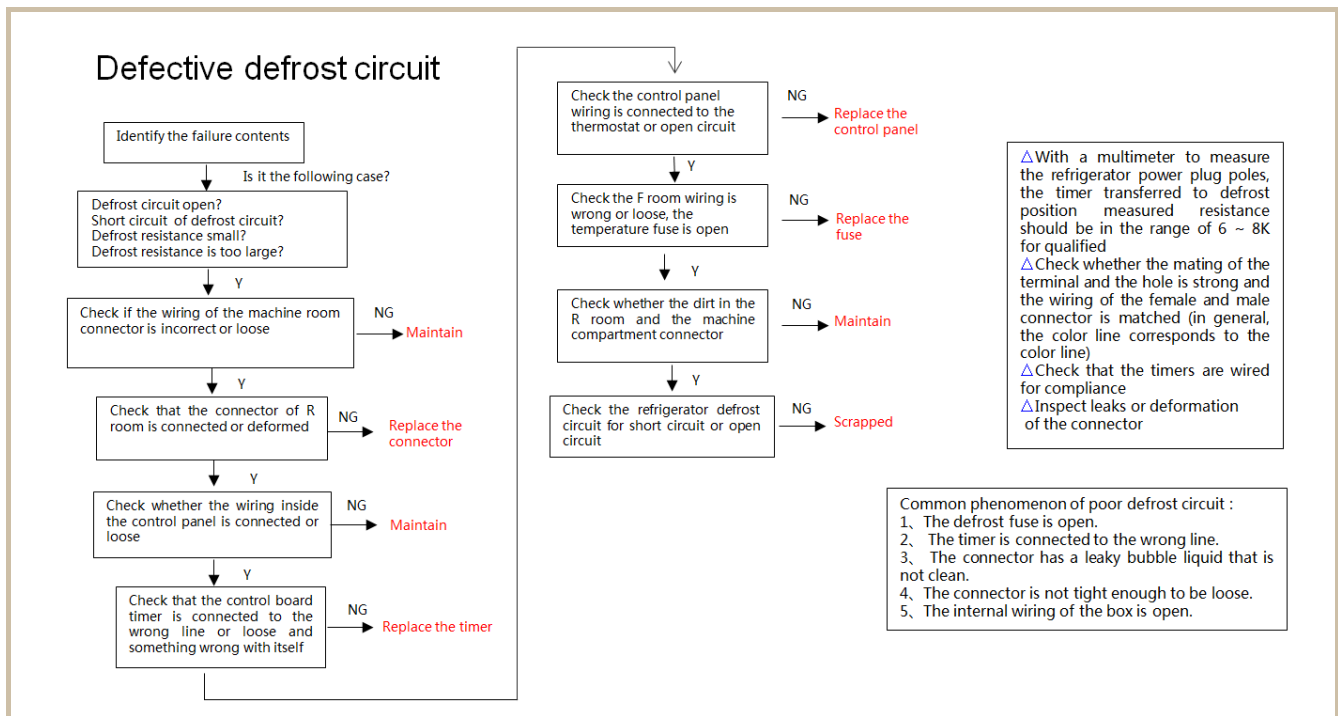
12.5 Light is not on



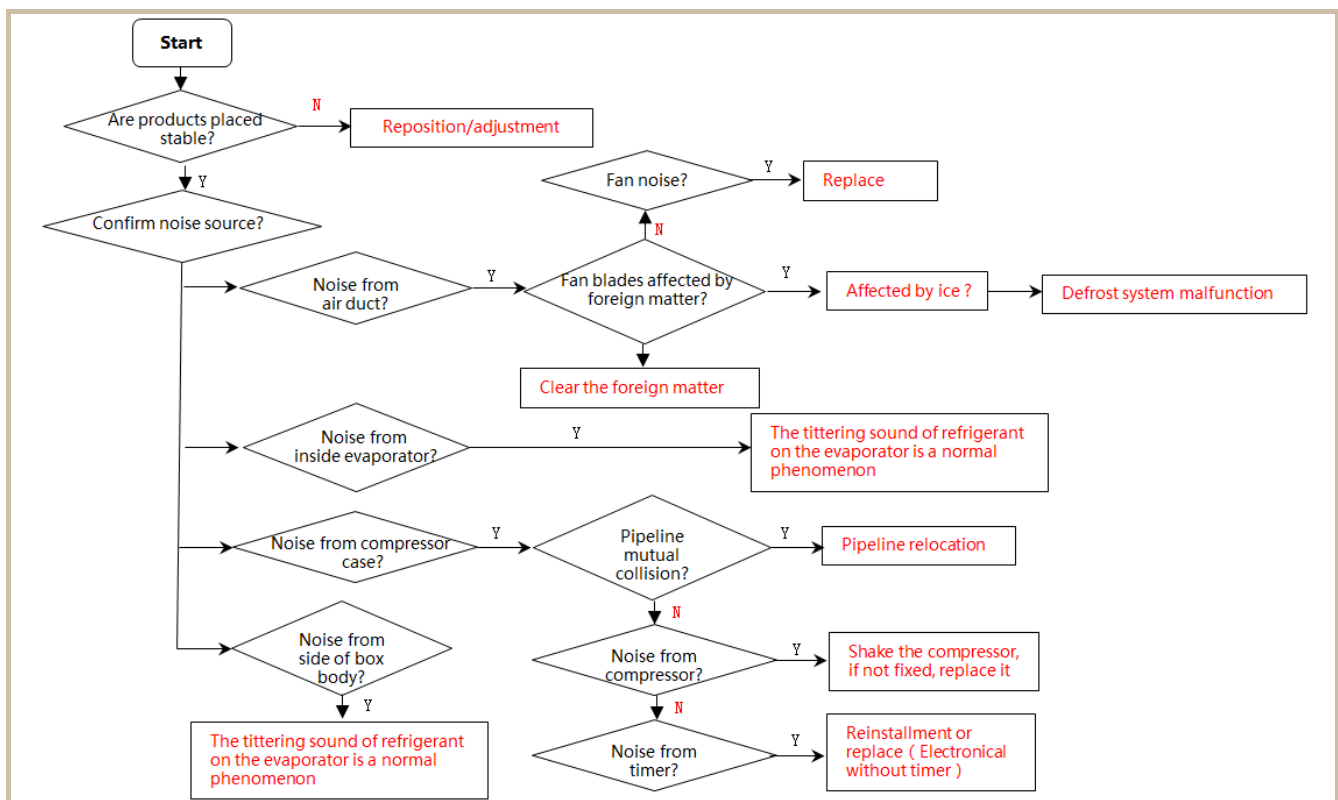
12.6 Fan failure



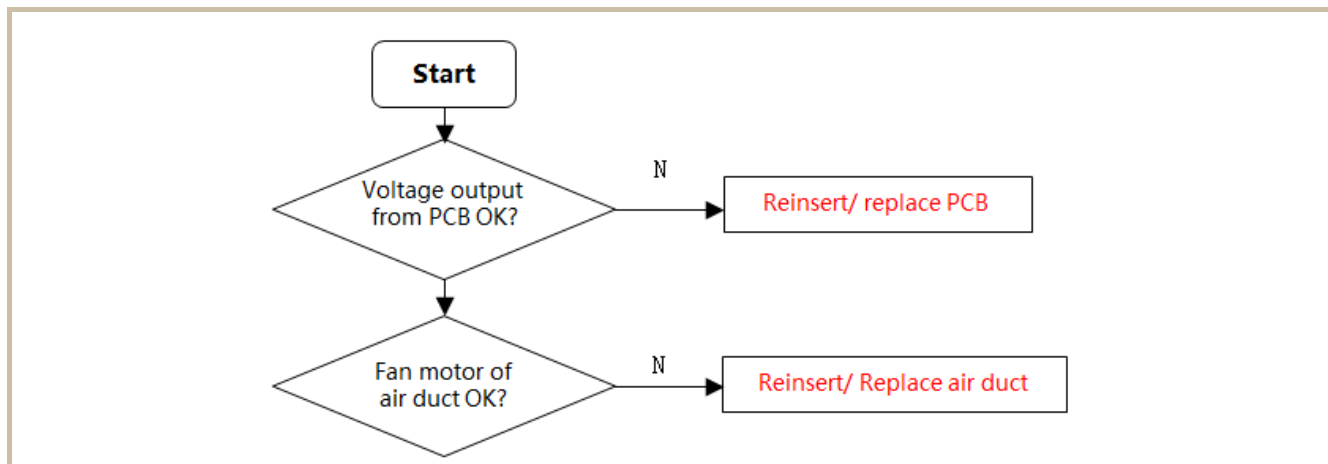
12.7 Defective defrost circuit



12.8 Noise



12.9 Air duct not operated(electronically)



12.10 ice maker not make ice

Keep pressing the “FRE” and “Energy Saving” button for 3s, enter/exit EYE.

Enter EYE, the display digital of ice machine mode will blink, and water gets into ice machine after turning over the ice twice



The symbol on the product or its packaging indicates that this product must not be disposed of with your other household waste. Instead, it is your responsibility to dispose of your waste equipment by handing it over to a designated collection point for the recycling of waste electrical and electronic equipment. The separate collection and recycling of your waste equipment at the time of disposal will help conserve natural resources and ensure that it is recycled in a manner that protects human health and the environment. For more information about where you can drop off your waste for recycling, please contact your local authority, or where you purchased your product.